The Language of Gesture: Testing the Multimodality-for-Compensation Account with Head and Hand Emblems of Affirmation and Negation

Samuel Sonntag[1], Ian Grant Mackenzie[1], Emanuel Schütt[1], Katarina Sivec[1], Sara Oguz[2], Merle Weicker[2], and Carolin Dudschig[1]

[1]Department of Psychology, University of Tübingen

[2]Goethe-University Frankfurt, Institute for Psycholinguistics

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Author Note

Correspondence concerning this article should be addressed to Samuel Sonntag, Department of Psychology, Research Group Language and Cognition, University of Tübingen, Schleichstraße 4, 72076 Tübingen, Germany. Email: samuel.sonntag@uni-tuebingen.de

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Abstract

Language is inherently multimodal in nature. However, the extent to which verbal and gestural information are integrated to form a coherent meaning is still not fully understood. While previous research has focused on iconic gestures, the present study investigates the integration of emblematic gestures—culturally specific gestures with direct verbal translations. Specifically, we examine how basic verbal and gestural affirmative and negative responses to polar questions are processed, both in isolation and in combination. Negation processing is known to be particularly demanding, making it an ideal test case for the *multimodality-for-compensation* hypothesis, which posits that multimodal cues are especially beneficial under increased processing difficulty. Across three experiments, participants responded to simple yes–no questions answered by either unimodal, compatible or incompatible multimodal responses ("yes"/"no" in combination with head nods/shakes and thumbs up/down gestures). The results demonstrated consistent integration effects between verbal and gestural information, with stronger integration effects for affirmative compared to negative responses. Implications of these findings for theories of multimodal language processing are discussed.

Public Significance Statement

Human language is inherently multimodal, combining speech with gestures, facial expressions, and other cues. Understanding how these different modes interact is key to uncovering how we comprehend language. This study explores whether emblematic gestures—culturally learned gestures like head shakes, nods, and thumbs up/down—can support the processing of negation, a universal but cognitively challenging feature of language. Specifically, the integration of head-shake/nod and thumbs-up/down gestures was investigated. While these gestures were found to be automatically integrated during comprehension, they did not fully ease the processing challenges associated with negation. Nevertheless, gestures may help support understanding when speech alone is more difficult to process. These findings shed light

on the limits of multimodality in language processing and highlight the need to further examine how specific gesture types influence comprehension.

Keywords: gestures, negation, multimodal communication, conflict

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Multimodality plays an important role in language processing. Although some researchers argue that language is inherently multimodal (i.e., consisting of visual and auditory signals, see Holler & Levinson, 2019), the specific function of multimodality remains uncertain. In the present study, we investigate whether multimodality is particularly advantageous in cognitively demanding contexts, using negation as a test case. Negation is a central aspect of human language. Surprisingly, despite the widespread use of negation in everyday communication, experimental studies on negation processing consistently report increased processing difficulties (for an overview, see Dudschig et al., 2019; Kaup & Dudschig, 2020). Also, the factors that facilitate the processing of negation remain a matter of debate (Orenes, 2021). While previous research has primarily examined the role of pragmatic licensing and predictability in negation processing (e.g., Nieuwland & Kuperberg, 2008), the present study investigates whether multimodal cues—specifically, speech in combination with head or hand gestures — can facilitate negation processing in responses to polar yes–no questions.

Multimodality

Human face-to-face communication involves more than the sequential interpretation of spoken words; it also requires the simultaneous processing of multimodal signals, such as gaze, facial expressions, and gestures that accompany speech. As an inherent part of communication, this is equally true for the processing of negation and affirmation, which can be expressed in various non-verbal forms, ranging from facial gestures like the "not-face" (Benitez-Quiroz et al., 2016; Schütt et al., 2024) to other expressions such as a head-shake/nod or a thumbs-down/-up gesture and many more (Beaupoil-Hourdel et al., 2016; Cooperrider, 2019; Harrison, 2014; Morgenstern et al., 2018; S. Wu et al., 2025).

While it is clear that both verbal and non-verbal aspects are essential to everyday communication, the precise role of non-verbal information is a topic of ongoing debate (Arbona et al., 2024; Demir-Lira & Göksun, 2024; Kita & Emmorey, 2023; Ter Bekke et al., 2025). Traditional views of the role of gesture in communication focus on gesture as a facilitator, but not as an essential or core aspect of language (e.g., lexical retrieval hypothesis; Krauss & Hadar, 1999). On the one side, gestures might facilitate production (see Bavelas et al., 2008, for a detailed summary) and help the speaker to organize information before verbalizing it (e.g., *information packing hypothesis*; Alibali et al., 2000; Kita, 2000). In this traditional view, gestures might also play a role for comprehension and speakers might produce gestures in order to aid or facilitate comprehension (e.g., Galati & Brennan, 2014). Again, this suggests that gestures function more as by-products of speech rather than being part of core linguistic processes.

In contrast are frameworks that regard verbal and non-verbal information as an inherent part of communication, forming a unified system (e.g., *integrated-systems hypothesis*; Kelly, Özyürek, & Maris, 2010; McNeill, 1992). McNeill (1992, 2005) suggested that understanding gestures is part of understanding language and thought. In this view, gestures play a fundamental role in language processing (McNeill, 1992)—for example, by reflecting the sensorimotor processes involved in thinking and speaking (Hostetter & Alibali, 2008) or as an evolutionary precursor to language (gesture-first hypothesis; e.g., Corballis, 2012). A similar view is put forward by Kelly, Özyürek, and Maris (2010), who report bidirectional influences of gesture and speech on comprehension. Indeed, brain regions responsible for both gesture and language production are closely linked (Bates & Dick, 2002; Skipper et al., 2007; Willems & Hagoort, 2007; Willems et al., 2007), suggesting that gestures and language evolved together. Alternatively, gestures can be seen as a bridge between pre-linguistic communication and the complex verbal languages that developed over time (Sterelny, 2012). Given the rather large amount of evidence pointing towards a close interconnection between language and gesture—these unified accounts are nowadays the predominant ones (Holler & Levinson, 2019).

Evidence for a unified language system has also been taken from studies showing the influence gestures can have on language processing. Empirical studies in this area are typically behavioral, neurophysiological, or developmental in nature, or a combination of these (Austin et al., 2014; Fabbri-Destro et al., 2015; Gunter & Bach, 2004; Holle & Gunter, 2007; Kelly, Creigh, & Bartolotti, 2010; Kelly et al., 2004, 2007; Krason et al., 2022; Y. C. Wu & Coulson, 2007a, 2007b). In these studies, the interaction between verbal and non-verbal information is typically supported in two ways: (1) through response speed effects, where the congruence between verbal and non-verbal information facilitates processing even if fully task-irrelevant (Kelly, Creigh, & Bartolotti, 2010; Kelly, Özyürek, & Maris, 2010), and (2) by the influence of non-verbal information on neurolinguistic networks closely associated with language processing, such as activation or interactions with the N400 complex—again often reported even if the information was fully task-irrelevant (Fabbri-Destro et al., 2015; Kelly, Creigh, & Bartolotti, 2010; Kelly et al., 2007).

Given the extensive research showing that verbal and non-verbal information directly interact during comprehension, both at the neural and behavioral level, the literature has shifted towards a more precise investigation of the function of gestures in comprehension. Specifically, the literature has focused on understanding in which context gestures are particularly helpful or the particular types of gestures that help comprehension most (Dargue et al., 2019). Interestingly, the positive influence of multimodal input has often been observed in contexts of language processing difficulties, such as ambiguity (Holle & Gunter, 2007; Holler & Beattie, 2003; Kidd & Holler, 2009), sound-related challenges (Drijvers & Özyürek, 2017; Krason et al., 2022), and in child comprehenders (Austin et al., 2014; McNeil et al., 2000). Taken together, these findings suggest that gestures may particularly aid when language processing becomes difficult—we will refer to this account as the *multimodality for compensation* account. The following section outlines some key empirical findings that have informed this theoretical proposal.

With regard to the help of multimodality for ambiguity resolution, for instance, Holle and Gunter (2007) demonstrated that during comprehension of ambiguous sentence beginnings (e.g., “She controlled the ball…”; ball referring either to a game or a dance), accompanying gestures facilitated semantic disambiguation. This effect was evidenced by EEG N400 activity (Kutas & Hillyard, 1984) following the onset of the disambiguating word (e.g., game or dance) later in the sentence continuation. In this view gestures are routinely used, somewhat subconsciously to facilitate or disambiguate speech comprehension (Holle & Gunter, 2007), thereby serving a meaningful purpose in reducing cognitive strain. Another line of evidence originates from studies showing that in the presence of sound-related challenges (e.g., degraded speech signal) to comprehension gestural input is particularly beneficial (Drijvers & Özyürek, 2017). For example, Drijvers and Özyürek (2017) used different levels of noise-vocoding and showed that the more difficult the auditory signal was to perceive, the more multimodal cues (iconic gestures and lip movements) facilitated comprehension (see also, Krason et al., 2022). A subsequent MEG study demonstrated that frequency modulations across extended language networks mediate attentional allocation to gestural information depending on speech comprehensibility, thereby indicating a functional interplay in the integration of multimodal input streams depending on processing difficulty (Drijvers et al., 2018; see also Holle et al., 2010). Similarly, it has been shown that hearing-impaired individuals consistently rely on gestural information, whereas non-hearing-impaired individuals do so primarily under conditions of externally induced comprehension difficulty (e.g., background noise). This suggests that both internal and external factors contributing to comprehension challenges enhance the reliance on gestures during communication (Obermeier et al., 2012). Moreover, there is evidence suggesting that gestures may play a more prominent role in comprehension among less experienced comprehenders, such as children, compared to experienced adult comprehenders; however, the empirical findings on this issue are mixed (Austin et al., 2014; Church et al., 2000; Dargue et al., 2019; Hostetter, 2011; Kartalkanat & Göksun, 2020; McNeil et al., 2000). Similarly, it has been shown that gestures accompanying speech help the comprehension of non-native speakers (Billot-Vasquez et al., 2020). Also, in second language learning, it has been shown that speech accompanying gestures particularly help learning for difficult words—and less so easy words—in adulthood (Gluhareva & Prieto, 2017).

Taken together, these findings suggest that multimodal input is especially helpful in situations with increased processing demands and when language processing becomes difficult, such as the processing of ambiguity, developing language systems, or poor input signals—driven by external (noise covering) or internal (hearing impairment) factors. Thus, multimodality might act as a compensatory mechanism to overcome cognitive challenges. Although the findings summarized above point towards a *multimodality for compensation account*, such an account has—to our knowledge—not yet been directly tested with regard to difficulty arising from linguistic structures—besides ambiguity resolution. Here we aim to directly test whether multimodal input serves a compensatory function overcoming cognitive challenges particularly in complex or demanding language processing contexts. Specifically, we will test whether multimodality particularly facilitates negation processing, an operator often associated with processing difficulties (e.g., Kaup & Dudschig, 2020).

Negation

Language without negation is inconceivable (Horn, 2001; Zeijlstra, 2022). Nevertheless, processing difficulties associated with negation have been observed across a wide range of negation phenomena (for an overview, see Farshchi et al., 2019; Kaup & Dudschig, 2020; Wales & Grieve, 1969). Early behavioral studies often employed verification paradigms, in which participants evaluated sentences against a presented picture (e.g., Carpenter & Just, 1975; Chase & Clark, 1971) or background knowledge (Wason, 1961; Wason & Jones, 1963). Other paradigms examined the influence of negation versus affirmation on sentence completion tasks (Donaldson, 1970; Wason, 1959, 1961), the ability to draw inferences from negated and affirmative sentences (e.g., Carpenter, 1973; Just & Clark, 1973), and the capacity to follow simple instructions (e.g., Dudschig & Kaup, 2018). Across these paradigms, negation consistently led to increased processing difficulties, as indicated by longer reaction times (RTs) and/or higher error rates (ERs).

Several explanations have been proposed for the processing difficulties introduced by negation. One widely debated explanation is that negation imposes a cognitive burden due to its inherently negative connotation. This idea stems from strong associations in early life with prohibition and control, as well as its connection to volitional aspects such as dislike or rejection (Dimroth, 2010). Another potential explanation is provided by models that posit the processing of negation involves two steps: first activating and then suppressing the to-be-negated information (Clark & Chase, 1972; Dudschig & Kaup, 2018, 2020), as well as accounts that focus on inhibition processes specifically (e.g., Beltrán et al., 2021). Another explanation first proposed by Wason (1965), suggests that in experiments, negation is often presented in highly artificial contexts—introducing negation without any contextual support or pragmatic licensing. In contrast, negation in everyday language typically serves to highlight deviations from expectations or to correct false propositions (see Givón, 1978, 1979). However, findings have been controversial, reflecting the ongoing debate in the field. While some studies have shown that pragmatic licensing can mitigate or even overcome negation difficulties (e.g., Nieuwland & Kuperberg, 2008; Orenes et al., 2016), others have found that affirmative control conditions benefit similarly from licensing or context, suggesting that negation difficulty persists (Darley et al., 2020; Rück et al., 2022). In summary, pragmatic factors play a central role in investigations of negation processing and can reduce some processing difficulties; however, other challenges associated with negation processing appear to persist.

Beyond processing difficulties in sentential negation, responses to basic yes–no decisions are typically slower when the answer is negative (e.g., Dudschig et al., 2023). Polar questions are also interesting in the context of pragmatics (see above) as they can easily be incorporated into naturalistic scenarios, which may help overcome potential pragmatic issues (for studies with children see, Austin et al., 2014; Feiman et al., 2017). Previous studies have shown that specific non-linguistic contextual features (e.g., colors, shapes, facial expressions), facilitate negation under certain conditions (Dudschig et al., 2023; Schütt et al., 2024), specifically in responses to polar questions—indicating that non-linguistic features can indeed facilitate negative answers (see also Benitez-Quiroz et al., 2016). However, the relative disadvantage compared to affirmation remained. In the present study, we aimed to investigate whether these processing difficulties in comprehending answers to polar questions can be overcome or reduced through multimodal language input; specifically, input that is more directly related to actual language processing (i.e., gestures) than the previously studied color and shape cues. Additionally, we adopt a more naturalistic, conversational paradigm for polar questions, building on approaches previously used with children (see Austin et al., 2014; Feiman et al., 2017).

The present paradigm

Negation is particularly interesting for studying the role of multimodality in language processing. Across cultures, there seem to be more gestures for negation than for affirmation (Gawne, 2021; Harrison, 2024), perhaps because negation requires more explicit signaling to prevent misunderstanding. In the present study, we adapted the paradigms from Austin et al. (2014) and Feiman et al. (2017) to investigate multimodal negation processing. In these studies, comprehension of negation as an answer to polar questions in children was investigated across different types of verbal negation (Feiman et al., 2017) and, additionally, for gestural negation (Austin et al., 2014). Notably, these studies employed a naturalistic setup that embedded negation within a context. In the original paradigms, children had to locate a toy in one of two visible opaque containers. The experimenter would give the children an answer to a question regarding the location of the toy ("Is the toy in there?"), with the answer consisting of head-nod/-shakes, "yes"/"no", and "It's (not) in the bucket". By the age of two, children could process negation and respond to these questions at above-chance levels.

Participants were instructed to determine the location of an imagined ball. At the beginning of each trial, participants were asked, "Is the ball in the blue/green box?" They then watched a video providing the correct answer by affirming (positive) or rejecting (negative) the question. In Experiment 1, all videos contained a verbal utterance accompanied by a gesture, with each indicating a possible answer (yes or no).

The gestures and verbal utterances could either be compatible (e.g., yes-verbal and yes-gesture) or incompatible (e.g., yes-verbal and no-gesture). In Experiment 2, only head gestures were included, and in Experiment 3, only hand gestures were included. Furthermore, the design was adapted for both Experiments 2 and 3 to also include a unimodal condition (e.g., gesture-only or speech-only) for comparison. Participants were explicitly instructed to focus on either the verbal or gestural modality and to ignore the other. To examine the generalizability across different gesture types, both head- (nod vs. shake) and hand gestures (thumb up vs. down) were investigated. These gesture types were selected based on the findings of Noah et al. (2017), who demonstrated neural integration patterns consistent with conflict processing for both head and hand gestures.

We predict the following outcomes. First, since negation is generally more challenging to process than affirmation, we expect response times to be slower in negation trials. Second, if gestural and verbal information are integrated directly—as suggested by the view that gestures are an inherent part of language (e.g., *integrated-systems hypothesis*, Kelly, Özyürek, & Maris, 2010; see also McNeill, 1992)—participants should respond more quickly when the two modalities provide compatible information compared to when they provide incompatible information. Third, and crucially, if gestures are particularly beneficial for processing difficult linguistic structures—as suggested by a *multimodality for compensation account* (e.g., Holle & Gunter, 2007; Krason et al., 2022)—we expect negation trials to show the greatest benefit from matching gestures.

Experiment 1

Transparency and Openness

Data were analyzed using *R*, version 4.4.2 (R Core Team, 2021), using the packages *tidyverse*, version 2.0.0 (Wickham et al., 2019), *psychReport*, version 3.0.2 (Mackenzie & Dudschig, 2022), *afex*, version 1.4-1 (Singmann et al., 2024) and *emmeans*, version 1.10.0 (Lenth, 2024). Plots were generated using *ggpubr*, version 0.6.0.999 (Kassambara, 2023) and *ggrain*, version 0.0.4 (Allen et al., 2021). Tables were generated with the help of *kableExtra*, version 1.4.0 (Zhu, 2024). All data and analysis code (for all three experiments) are available as additional online material (https://doi.org/10.5281/zenodo.17573533). Stimulus material is available for the scientific community from the first author upon valid request. Data collection was completed between December 20, 2023 and February 13, 2024. The Experiment was preregistered (https://aspredicted.org/rwfx-4wq4.pdf). In line with the preregistration, all statistical tests were conducted using frequentist analysis and stayed consistent within this approach (Dienes, 2024). If there were any departures from the pre-registration these are detailed in the text.

Method

Participants

A total of 155 German native speaking participants ($M_{\text{age}} = 30.55$ years, $SD_{\text{age}} = 10.06$, *range* = 18–60, 75 women, 78 men, 2 non-binary individuals) took part in this study. In total, 134 participants stated that they were right-handed (21 left-handed). In absence of directly comparable prior studies, the sample size in the preregistration was determined based on the estimates provided by Brysbaert (2019). For a 2x2 within-subjects design and a medium effect size of $d = 0.4$, a minimum of 145 participants was needed to achieve a statistical power of at least .90. Due to online data collection we collected 10 more participants than preregistered [1]. Due to an overall ER > 30%, a total of 9 participants were excluded from all analyses. All studies were approved by a local ethics committee (Ethics Proposal No. 2021_0505_226) and all participants gave informed consent. The participation was initially handled through the University of Tübingen *Sona System* (https://www.sona-systems.com), where participants received course credit. Because the full sample size was not reached, a second part of data collection was handled through Prolific (www.prolific.com), where participants were reimbursed with 9 GBP.

[1] When running the analysis without these 10 additional participants, the overall result pattern was consistent. We thus chose to report the full set of recorded participants.

Stimuli and Apparatus

All experiments reported in the current study were conducted online using the JavaScript library *jsPsych*, version 7.2.3 (De Leeuw et al., 2023). The experiments were hosted on a local web server of the University of Tübingen. The start of the experiment involved a calibration routine to control stimulus sizes. Here, participants had to resize a presented box using their mouse until it roughly matched a standard credit/id card in size. This routine was accompanied by a screen-check that ensured a minimum screen resolution of approximately 960 × 720 pixels. All stimuli were presented on a non-transparent white background within a canvas with a solid black frame. A centrally presented black plus sign served as a fixation cross. Each trial involved the presentation of a question ("Is the ball in the blue / green box?" / "Ist der Ball in der blauen/grünen Box?"), the presentation of a video and the presentation of a blue and green box in a randomly assigned bottom corner (left vs. right side) each (see Figure 1).

The videos were filmed using an iPhone 8 Plus—in 1920 x 1080 (full HD) resolution at 60 frames per second—in a sound attenuated room. Each video consisted of the presentation of a gesture (either affirmative or negative) and an utterance (either *yes* or *no*)—all videos were cut to 2000 ms length using Final Cut Pro (Apple Inc., 2023). The resulting possible four combinations of gestures and utterances were recorded for both a head (either *head shake* or *nod*) and a thumb gesture (*thumbs up* or *thumbs down*). This resulted in 16 videos in total, eight with a female and eight with a male actor. The two actors maintained a neutral demeanor and facial expression throughout all videos (Wood et al., 2019). The visible portion of the video captured the subjects from the mid-torso to above the head (see Figure 1). We selected this visible portion of the videos to maximize control over the gesture timing, including onset and to minimize potential differences between conditions that could arise from factors other than the critical variables under investigation (e.g., body posture etc.). While this setup aligns with certain real-life situations (e.g., video calls or communication when sitting at a table), it does not reflect a typical face-to-face interaction in which the whole body is visible (Obermeier et al., 2011). Each video started with them posing in a neutral position that did not allow for any conclusions to be drawn as to which gesture or speech followed (see Figure 1)— also excluding preparatory or extraction phases for gestures performances (Willems et al., 2007). Thus, it was avoided that information from the gesture was revealed before the relevant stroke or head movement begun. Previous studies indicated that gesture-speech integration would take place in the time frame from at least -200 to 120 ms synchrony (Obermeier & Gunter, 2015). Although studies in naturalistic production settings have shown that certain types of gestures (i.e. iconic, metaphoric, deictic; unclear for emblematic gestures) often precede speech (Ter Bekke et al., 2024b), we opted for temporal synchrony between gesture and speech to ensure that any differences in reaction times could not be attributed to timing differences in the stimuli. Thus, in the present study the synchrony of speech and gesture onset was highly controlled and it was ensured that gesture and verbal onset were temporally aligned. First, video recording took place until speech and gesture onset were perceived in synchrony. In a second step, synchrony of onset was checked, and if required, minimal edits were made using Final Cut Pro (Apple Inc., 2023). As a result, there were no systematic variations in onset across any experimental conditions (see Table D1 for detailed timing information). In order to ensure videos were perceived as naturalistic, six additional participants (M_{age} = 37.83 years, SD_{age} = 20.85, 4 women, 2 men) rated the videos on a scale from 1 (artificial) to 7 (naturalistic). Results showed that videos were perceived as rather naturalistic in the compatible conditions (head gesture: M = 6.46, SD = 0.78; hand gesture: M = 6.42, SD = 0.72). Since timing was controlled equivalently across the incompatible conditions, we assumed that these would appear comparably naturalistic, even if unlikely to occur spontaneously. It was also ensured that no systematic variation in loudness was confounded with any of our experimental factors using the tool *ffmpeg*, version 7.1.1 (Tomar, 2006). The peak and integrated loudness values by experimental factors are included in the appendix (see Table D2).

Procedure

Participants took part remotely using private computers. Participation with either phones or tablets was not possible as a keyboard was required for responding.

Participants were asked to complete the experiment in a quiet environment. The main task consisted of answering previously posed questions using information presented within the video. Each trial began with the presentation of a fixation cross for 1000 ms, followed by the presentation of the current question and the two colored boxes for 2500 ms (see Figure 1). The questions posed asked whether a ball was in the green ("Ist der Ball in der grünen Box?" / "Is the ball in the green box?") or in the blue box ("Ist der Ball in der blauen Box?" / "Is the ball in the blue box?"). The blue and the green box appeared at the bottom of the screen with the location—which box appeared on the left and which on the right side— varying randomly from trial to trial. This was followed by the presentation of a video with a duration of 2000 ms informing participants about the correct location of the ball (yes vs. no response to the previously posed question). Participants were instructed which modality (verbal vs. gestural) they had to respond to at the beginning of the experiment and in the middle of the experiment when the dimension was switched. The order (gesture-first vs. verbal-first) was balanced across participants (half of the participants started with gesture-first, the other half with verbal-first)[2]. Participants responded using the "F" and "J" keys. "F" (being the left key) always corresponded to the box presented on the left and "J" (being the right key) corresponded to the box on the right side—reducing the need for memorizing an arbitrary response mapping. If participants had not responded until the end of the video, the video disappeared and only the two colored boxes remained on the screen until a response was submitted. Participants received written feedback (1000 ms duration) in the center of the screen at the end of each trial ("Richtig!" / "Correct!" or "Falsch!" / "Incorrect!"). Each trial was followed by a blank inter-trial-interval of 1000 ms. The participants performed six experimental blocks with 64 trials each—all conditions were balanced within each block and each video was shown four times. Participants performed a practice block containing 10 trials before the first experimental block; the

[2] In order to assure that this switch in modality did not change any of our relevant result patterns, we conducted an additional analyses and added the factor *half* to our analyses. We did find a speed-up that was consistent with a practice effect; The compatibility effect was not significantly influenced by the modality switch in any of our experiments. These analyses can be found in the Zenodo repository (https://doi.org/10.5281/zenodo.17573533).

same was true for the fourth block, where the relevant dimension was changed. All blocks were separated by a self-paced break during which participants were presented with feedback regarding their average performance (RTs and ERs) within the previous block. On average, the experiment took approximately 53 minutes to complete.

Figure 1
Trial Sequence Experiment 1

Design

The dependent variables were RT and ER. The independent variables were in line with the preregistration and included response polarity (*negation* vs. *affirmation*) and the compatibility of gestural and verbal information (*compatible* vs. *incompatible*). Thus resulting in a 2x2 within-subjects design.

Results

All main analyses were conducted using repeated-measures ANOVAs. Follow-up analyses were conducted using the R-package *emmeans*, version 1.10.0 (Lenth, 2024) applying Bonferroni corrections to control for multiple comparisons[3]. The analysis was in line with the preregistration unless stated otherwise. Extreme RT outliers (< 200 ms or > 5000 ms) were excluded from all analyses. This amounted to 0.18% of the data. Additionally, all erroneous trials were removed from the RT analyses (6.09% of all trials).

Main Analysis: Compatibility x Polarity

Analyses were conducted using 2x2 repeated-measures ANOVAs. For the RTs in line with the first hypothesis, there was a main effect of polarity with shorter RTs for affirmative (M = 1054 ms, SEM = 16) compared to negated (M = 1140 ms, SEM = 19) trials, $F(1, 145) = 244.16$, $p < .001$, $\eta_p^2 = 0.63$. In line with the second hypothesis, there was a main effect of compatibility with responses being faster for compatible (M = 1070 ms, SEM = 17) compared to incompatible conditions (M = 1124 ms, SEM = 18), $F(1, 145) = 275.44$, $p < .001$, $\eta_p^2 = 0.66$.

The interaction between compatibility and polarity was also significant, $F(1, 145) = 131.41$, $p < .001$, $\eta_p^2 = 0.48$ (see Figure 2, A). However, this effect arose because, contrary to the *multimodality for compensation* account, compatibility was specifically beneficial in the case of affirmation. Indeed, follow-up analyses revealed that this interaction was due to a significantly larger difference between compatible and incompatible trials in the affirmative (M_D = 89 ms, SEM = 5), $t(145) = 17.97$, $p < .001$, $d_z = 1.49$, compared to the negated (M_D = 20 ms, SEM = 4) condition, $t(145) = 5.04$, $p < .001$, $d_z = 0.42$[4].

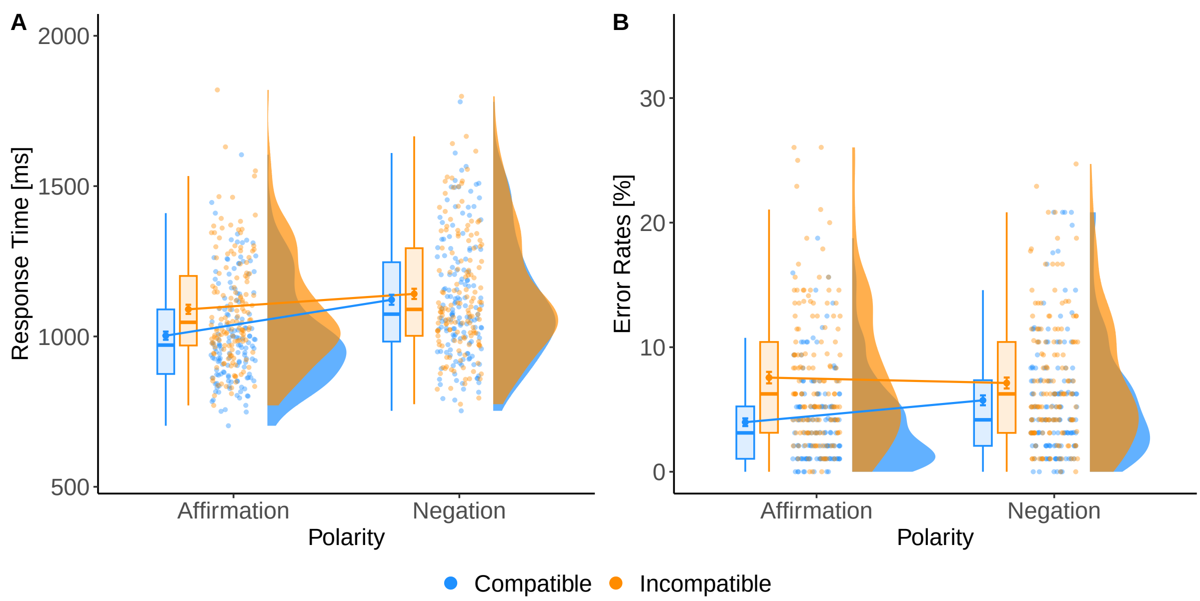
[3] No specification regarding the adjustment for multiple comparisons was made in the preregistration, thus we opted for the conservative Bonferroni adjustment.

[4] We have double checked that this interaction was not just driven by larger compatibility effects in faster RTs. If anything, compatibility effects increased for slower RTs in all experiments; These analyses can be found on the Zenodo repository (https://doi.org/10.5281/zenodo.17573533).

The analysis of ERs mainly matched the reported RT analysis pattern. Participants committed fewer errors for affirmative (M = 5.77%, SEM = 0.35) compared to negated (M = 6.43% SEM = 0.39) trials, $F(1, 145) = 5.88$, $p = .017$, $\eta_p^2 = 0.04$. Error rates were also lower for compatible (M = 4.86 %, SEM = 0.32) than for incompatible (M = 7.34%, SEM = 0.41) stimuli, $F(1, 145) = 99.52$, $p < .001$, $\eta_p^2 = 0.41$. The interaction between compatibility and polarity was significant, $F(1, 145) = 22.96$, $p < .001$, $\eta_p^2 = 0.14$ (see Figure 2, B). Follow-up analyses revealed a significantly larger difference between compatible and incompatible stimuli in the affirmative (M_D = 3.58%, SEM = 0.37), $t(145) = 9.75$, $p < .001$, $d_z = 0.81$, compared to the negated condition (M_D = 1.39%, SEM = 0.31), $t(145) = 4.54$, $p < .001$, $d_z = 0.38$.[5]

Figure 2

A: Reaction Times in milliseconds, B: Error Rates in % for Experiment 1



Note. Within both plots, grouping by Compatibility Condition in Color and Polarity on the X-Axis. Participants individual means with conditions are included within both plots, aggregated means (across participants) are also included within both plots. All aggregated means are surrounded by error bars indicating their respective standard error (+/- 1 SEM).

[5] To ensure that our findings were robust, we have computed all of our relevant analyses with log (logit for errors) transformed values. None of our relevant results (in any of the Experiments) were changed under these transformations. These analyses can be found on the Zenodo repository for all three Experiments (https://doi.org/10.5281/zenodo.17573533).

Additional Analysis: The Influence of Modality

This analysis, preregistered as an additional analysis, addressed the question whether the target modality impacted the core pattern of results regarding the *multimodality for compensation* account. Thus, in addition to the factors of polarity and compatibility, the factor modality was included in this analysis to examine whether response patterns differed when participants responded to verbal versus gestural information, resulting in a 2×2×2 repeated-measures analysis. The previously reported main effects and interactions were replicated. The full analysis can be found in the Appendix Tables, Table A3 and Table A4. Centrally, the three-way interaction between compatibility, polarity and modality did not reach significance, $F(1, 145) = 3.16$, $p = .077$, $\eta_p^2 = 0.02$. Thus, there was no evidence that the *polarity x compatibility* interaction was modified by modality—suggesting that with regard to the *multimodality for compensation* account, there was no evidence for a difference whether participants responded to verbal or gestural information. Within the ERs, no additional effects to the main analysis reached significance (see also Table A1 for means and Table A2 for statistical results).

Additional Analysis: The Influence of Gesture

This additional, non-preregistered, exploratory analysis allows for some more detailed insights regarding differences for the gesture types under investigation with regard to the *multimodality for compensation* account. Thus, in addition to the basic factors, the factor gesture type was included in this analysis in order to explore whether there were any differences in the result patterns for the head gestures versus the hand gestures—resulting in a 2x2x2 repeated measure ANOVA. The main effects and interactions mirrored the main analysis described above (for the full analysis see Appendix Table A5 and Table A6). The relevant comparison, the three-way interaction between compatibility, polarity and gesture, $F(1, 145) = 29.50$, $p < .001$, $\eta_p^2 = 0.17$ did reach significance. This was due to the compatibility x polarity interaction being more pronounced for the head gestures, $F(1, 145) = 112.11$, $p < .001$, $\eta_p^2 = 0.44$ than for the hand gestures, $F(1, 145) = 30.45$, $p < .001$, $\eta_p^2 = 0.17$ (see Table A1 for means).

Importantly, a consistent pattern was observed across both gesture types, namely, the compatibility effect was greater for affirmative, $t(145) = 17.00, p < .001, d_z = 1.41$ (head gesture), $t(145) = 10.55, p < .001, d_z = 0.87$ (hand gesture) than negated trials, $t(145) = 3.34, p = .001, d_z = 0.28$ (head gesture), $t(145) = 3.40, p < .001, d_z = 0.28$ (hand gesture).

Within the ERs, the three-way interaction between compatibility, polarity and gesture type also reached significance, $F(1, 145) = 4.44, p = .037, \eta_p^2 = 0.03$. Similar to the RT analysis, this was caused by a larger compatibility x polarity interaction for head gestures, $F(1, 145) = 23.08, p < .001, \eta_p^2 = 0.14$ than for hand gestures, $F(1, 145) = 4.60, p = .034, \eta_p^2 = 0.03$ (see Table A1 for means).

Discussion

The results showed the predicted deleterious effect of negation on both RTs and ERs. Responses were slower and more erroneous for negated information, present both when responses were required to verbal and to gestural information. Thus, the present study showed that the difficulties in processing negation are not limited to verbal information but are also present when responding to gestural information. Additionally, responses were slower and less accurate when gestural and verbal information were incompatible—this effect was again present both when responding to gestural or to verbal information. This suggests that both task-irrelevant verbal and task-irrelevant gestural information are directly integrated during the comprehension process—even when the information is fully task-irrelevant.

One of the central questions of the present experiment concerned whether multimodal input particularly helps with processing of more difficult linguistic input. Indeed, the results showed a significant interaction of polarity and the factor compatibility. However, this was due to the compatibility of multimodal information having a larger benefit in the affirmative compared to the negated condition. In other words, participants seemingly benefited from the presence of multimodality most within the condition that was less difficult to process. While there are findings that suggest that multimodal information doesn't always benefit processing to the same degree (see

Drijvers & Özyürek, 2020; Holle & Gunter, 2007; Kelly, Creigh, & Bartolotti, 2010; Krason et al., 2022), the present results suggested that difficulty alone does not account for the increased benefits of multimodality in certain situations.

Interestingly, an additional follow-up analysis showed that the central findings replicated across the two gesture types—head gestures and hand gestures. This is evidence that the present findings are indeed rather general effects for affirmative and negated emblematic gestures, and not limited to one specific type of gesture.

Experiment 2

Experiments 2 and 3 aimed to further corroborate the core findings from Experiment 1 by investigating the two gesture types separately, thereby avoiding potential carry-over effects from their random intermixing in Experiment 1. We also sought to replicate the finding that multimodality facilitated comprehension more for affirmative than for negated responses. In addition, a unimodal condition was introduced in Experiments 2 and 3 —either a unimodal verbal or gestural "yes" versus "no" response. This condition allows us to investigate the beneficial or possibly deleterious influence of multimodality relative to a unimodal condition which conveys less information value, but potentially imposes lower processing demands. If compatibility specifically facilitates processing, we expect the unimodal condition to be more difficult than the compatible condition. Conversely, if incompatibility hinders processing, we expect the unimodal condition to be easier. A mixture of both processes is also possible.

Transparency and Openness

The experimental setup was similar to Experiment 1; therefore, only aspects differing from Experiment 1 will be explicitly mentioned in the following section. Data collection was completed between October 12, 2024 and April 15, 2025. The experiment was preregistered (https://aspredicted.org/nvzn-hq83.pdf).

Method

Participants

We used the results from Experiment 1 to guide the planned sample size. Specifically, using the R-package *Superpower*, version 0.2.3 (Lakens & Caldwell, 2021), we simulated the number of participants required to achieve a power of .9, estimated from the cell means and standard deviations observed in Experiment 1. We used a fixed correlation value of .9 (NB, actual observed correlations between conditions within Exp 1. were approximately .95). As Experiments 2 and 3 separated gesture types, a conservative estimated sample size of 131 was used as a baseline. As preregistered, 150 German native speaking participants (M_{age} = 27.77, SD_{age} = 11.27, *range* = 18–75, 84 women, 62 men, 4 non-binary individuals) took part in this study to account for exclusions. In total, 131 participants stated that they were right-handed (19 left-handed). Due to an overall ER > 30%, a total of 12 participants were excluded from all analyses. With a final sample size of 138 participants after exclusions, no replacements had to be conducted.

Stimuli and Apparatus

Only the head gestures were investigated here. In addition to the previously implemented multimodal videos, the videos were also modified to create the unimodal conditions—in affirmative and negated form. Half of the unimodal videos were gesture-only videos whilst the other half were verbal-only. It was specifically ensured that the unimodal videos do not differ in timing parameters from the multimodal videos. In order to do so, all unimodal videos were created directly from the multimodal videos. This was achieved by means of muting all sounds in the gesture-only unimodal video (resulting in a female head-shake, female head-nod and male head-shake and head-nod video). In the verbal-only condition a freeze frame was implemented (again for all polarities and speakers). The result being a “unimodal” condition that conveyed relevant information within only one stimulus dimension, without any potential confounds in timing of gesture or voice onset.

Procedure

The procedure was largely identical to Experiment 1, except for the addition of the unimodal trials. In each block, 32 unimodal trials were randomly intermixed with 32 compatible and 32 incompatible trials. To account for this addition, the total length of the experiment was reduced by 2 blocks. Thus, the experiment consisted of 4 experimental blocks. Overall, this resulted in a total of 384 trials. The experimental duration across all participants was approximately 51 minutes.

Design

The design was adapted from Experiment 1 with the addition of the unimodal conditions. While originally planned to compare the unimodal condition separately to each compatibility condition—we deviated from this preregistered plan and incorporated it in the full design, resulting in a 2x3 repeated measures design with the factors polarity (affirmative vs. negative) and compatibility (unimodal vs. compatible vs. incompatible)[6].

Results

All main analyses were conducted using 2x3 repeated-measures ANOVAs. Follow-up analyses were conducted using the R-package *emmeans*, version 1.10.0 (Lenth, 2024) to adjust for multiple comparisons. If necessary, Greenhouse-Geisser corrections were applied to account for violations of sphericity. All departures from the preregistration are detailed in the text. Extreme RT outliers (< 200 ms or > 5000 ms) were excluded from all analyses. This amounted to 0.47% of the data. Additionally, all erroneous trials were removed for the RT analyses (6.08% of all trials). Overall results summary and ANOVA tables can be found in the Appendix (Section B) with corresponding analyses on Zenodo (https://doi.org/10.5281/zenodo.17573533)—for brevity we focus on the tests most relevant for the hypothesis in the following sections.

[6] This change was suggested by a reviewer and did not alter any of our central result patterns. The original preregistered analysis is also available on Zenodo (https://doi.org/10.5281/zenodo.17573533).

Main Analysis: Compatibility x Polarity

The results are displayed in Figure 3. As in Experiment 1, there was a main effect of polarity with shorter RTs for affirmative (M = 1066 ms, SEM = 24) than for negative trials (M = 1167 ms, SEM = 26), $F(1, 137) = 242.34$, $p < .001$, $\eta_p^2 = 0.64$. As in the previous experiment, there was also a significant main effect of compatibility, $F(1.80, 247.00) = 165.74$, $p < .001$, $\eta_p^2 = 0.55$, $\epsilon = 0.90$. Follow-up analyses revealed a significant difference between unimodal (M = 1062 ms, SEM = 23) and compatible trials (M = 1099 ms, SEM = 25), $t(137) = 5.19$, $p < .001$, $d_z = 0.44$, with unimodal trials eliciting the fastest responses. There was also a significant difference between compatible and incompatible trials (M = 1188, SEM = 27), $t(137) = 14.82$, $p < .001$, $d_z = 1.26$, with incompatible trials eliciting the slowest responses.

Importantly, there was also a significant interaction between compatibility and polarity on RTs, $F(1.97, 269.21) = 49.78$, $p < .001$, $\eta_p^2 = 0.27$, $\epsilon = 0.98$. Post-hoc analyses revealed that the interaction between compatibility and polarity was also significant when including compatible and incompatible trials only, $F(1, 137) = 89.25$, $p < .001$, $\eta_p^2 = 0.39$, replicating the results from Experiment 1. In detail, the effect of compatibility between just compatible and incompatible trials was larger on affirmative (M_D = 128 ms, SEM = 7; $t(137) = 17.45$, $p < .001$, $d_z = 1.49$), compared to negative trials (M_D = 50 ms, SEM = 7; $t(137) = 6.89$, $p < .001$, $d_z = 0.59$). Again being in line with the findings from Experiment 1. The interaction between compatibility and polarity was also significant between only compatible and unimodal trials, $F(1, 137) = 24.88$, $p < .001$, $\eta_p^2 = 0.15$. The difference between unimodal and compatible trials was smaller on affirmative (M_D = 18 ms, SEM = 7; $t(137) = 2.52$, $p = .025$, $d_z = 0.21$), compared to negative trials (M_D = 55 ms, SEM = 9; $t(137) = 6.40$, $p < .001$, $d_z = 0.54$). For full comparisons, see Table B1 and Table B2.

There was no evidence for a significant difference of ERs for affirmative (M = 5.87 %, SEM = 0.38) and negated trials (M = 6.31 %, SEM = 0.37), $F(1, 137) = 2.41$, $p = .123$, $\eta_p^2 = 0.02$. There was a significant effect of compatibility on ERs, $F(1.70, 232.59) = 90.41$, $p < .001$, $\eta_p^2 = 0.40$, $\epsilon = 0.85$. Follow-up analyses revealed no evidence for a significant difference between unimodal (M = 4.98 %, SEM = 0.35) and compatible trials (M = 5.04 %, SEM = 0.32), $t(137) = 0.27$, $p = 1.000$, $d_z = 0.02$. There was a significant difference between compatible and incompatible trials (M = 8.25 %, SEM = 0.47), $t(137) = 10.62$, $p < .001$, $d_z = 0.90$. There was also a significant interaction between compatibility and polarity on ERs, $F(1.84, 252.56) = 23.13$, $p < .001$, $\eta_p^2 = 0.14$, $\epsilon = 0.92$. Post-hoc analyses revealed no significant interaction between compatibility and polarity between unimodal and compatible trials only, $F(1, 137) = 0.77$, $p = 1.000$, $\eta_p^2 = 0.01$. There was no evidence for a significant difference between compatible and unimodal trials across affirmation (M_D = 0.14 %, SEM = 0.28; $t(137) = 0.48$, $p = 1.000$, $d_z = 0.04$) and negation (M_D = 0.25 %, SEM = 0.33; $t(137) = 0.77$, $p = .890$, $d_z = 0.07$). The interaction between compatibility and polarity between compatible and incompatible trials was significant, $F(1, 137) = 33.24$, $p < .001$, $\eta_p^2 = 0.20$. The difference between compatible and incompatible trials was larger for affirmation (M_D = 4.85 %, SEM = 0.46; $t(137) = 10.52$, $p < .001$, $d_z = 0.90$) compared to negation (M_D = 1.58 %, SEM = 0.36; $t(137) = 10.52$, $p < .001$, $d_z = 0.90$). Incompatible trials yielded higher ERs than compatible trials across conditions. For full comparisons, see Table B1 and Table B3.

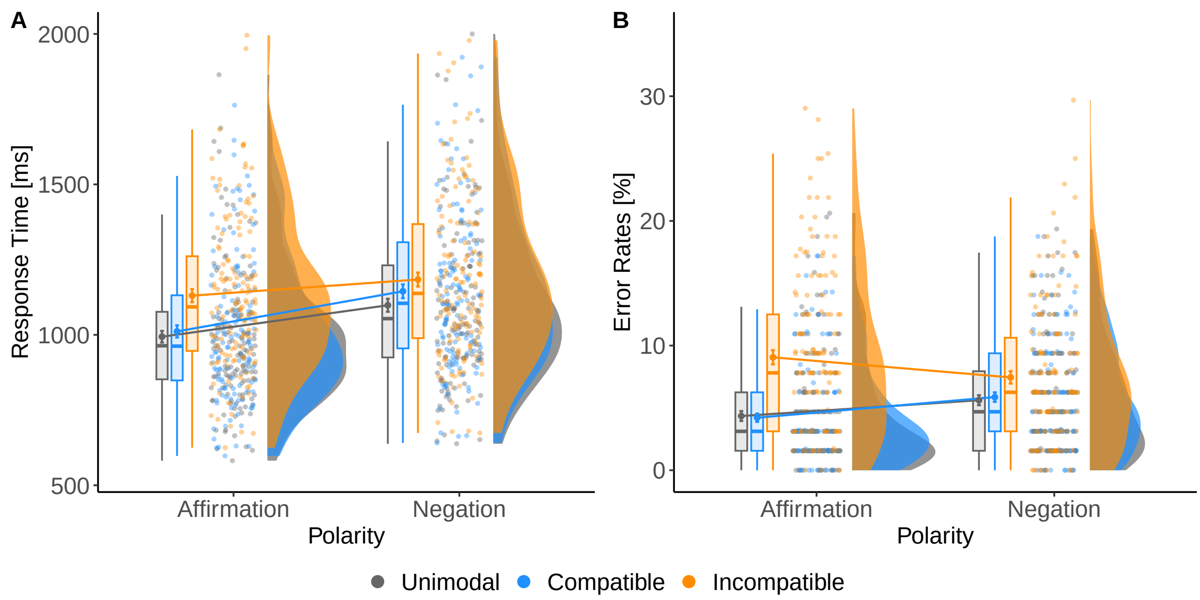
Additional Analysis: The Influence of Modality

As preregistered, additional analyses that included the factor modality were conducted for more detailed insights into the influence of participants responding to the verbal or gestural information with regard to the *multimodality for compensation* account. All analyses conducted in this subsection were (part of a) 2x2x3 repeated measures ANOVA.

Interactions and main effects reported in the main analysis were mirrored by the present results in both RTs and ERs. The full analyses can be found in Table B4 and Table B5. Critically, the three-way interaction between compatibility, polarity and modality did not reach significance, $F(1.73, 236.44) = 1.62$, $p = .202$, $\eta_p^2 = 0.01$, $\epsilon = 0.86$. Thus, in line with the findings from Experiment 1, there was no evidence that the

Figure 3

A: Reaction Times in milliseconds, B: Error Rates in % for Experiment 2



Note. Within both plots, grouping by Compatibility Condition in Color and Polarity on the X-Axis. Participants individual means with conditions are included within both plots, aggregated means (across participants) are also included within both plots. All aggregated means are surrounded by error bars indicating their respective standard error (+/- 1 SEM).

main results were significantly affected by whether participants responded to verbal or gestural information.

However, there was an interesting pattern revealed in this sub-analysis. There was an interaction between compatibility and modality, $F(1.77, 242.29) = 67.83$, $p <$ .001, $\eta_p^2 = 0.33$, $\epsilon = 0.88$. Looking into this interaction, when responding to verbal information—where the unimodal condition consisted only of speech— the incompatible condition (M = 1208 ms, SEM = 27) was the slowest, followed by the unimodal (M = 1133 ms, SEM = 24) condition (incompatible vs. unimodal: $t(137) = 9.50$, $p < .001$, d_z = 0.81), while the compatible (M = 1100 ms, SEM = 24) condition elicited the fastest responses (unimodal vs. compatible: $t(137) = 5.31$, $p < .001$, $d_z = 0.45$). This was not the case when gesture was the response-relevant dimension—and thus the unimodal condition consisted of only gestures. In this case, the incompatible condition was the slowest again (M = 1171 ms, SEM = 32), followed by the compatible condition (M = 1098 ms, SEM = 28) (incompatible vs. compatible: $t(137) = 8.97$, $p < .001$, $d_z = 0.76$), while the unimodal condition elicited the fastest responses (M = 993 ms, SEM = 26) (compatible vs. unimodal: $t(137) = 8.94$, $p < .001$, $d_z = 0.76$).[7]

Within the ERs, the critical three-way interaction between compatibility, modality and polarity did not reach significance ($F(1.88, 257.98) = 1.02$, $p = .358$, $\eta_p^2 = 0.01$, $\epsilon = 0.94$)—thus, there was no evidence that modality had a significant influence on the critical interaction.

Discussion

This experiment investigated head gestures only, with the addition of a unimodal condition. The core findings from Experiment 1 were replicated. Responses were slower and less accurate in the negated compared to the affirmative condition, indicating processing difficulties associated with negation. This negation-induced difficulty was present both when verbal or gestural information was task-relevant. Also, the compatible trials were faster to process than the incompatible trials in the overall analysis. Although at first glance, this seems to contradict some findings from the gesture literature (see Holler et al., 2018), it is consistent with some findings from the conflict processing literature—for example, in the case of the Eriksen Flanker task (see Eriksen & Eriksen, 1974; Yu et al., 2023, see also General Discussion for more details on this issue)—and follow up analysis showed that this pattern did indeed depend on the relevant task modality. Specifically, post-hoc analyses showed that the pattern that unimodal trials elicited the fastest responses was specifically driven by the condition were participants responded to gestures. In contrast, when responding to verbal stimuli only—participants were faster for compatible compared to unimodal trials—also in line with Holler and Levinson's (2019) proposal that multimodal language represents the default and most natural form of language processing. Taken together, these findings suggest that both task-irrelevant gestures and task-irrelevant verbal information directly impact processing. Most importantly, and in line with Experiment 1, particularly the

[7] It may be relevant to mention, despite the three-way interaction not reaching significance, compatible trials only elicited faster responses than unimodal trials in the affirmative speech condition, $t(137) = 7.88$, $p < .001$, $d_z = 0.67$. There was no evidence for a difference between compatible and unimodal trials in the negative speech condition, $t(137) = 0.93$, $p = 1.000$, $d_z = 0.08$. Thus, this result pattern is fully in line with the idea that compatibility is specifically helpful in affirmative compared to negated trials.

affirmative condition benefited from compatibility of gestural and verbal information, suggesting that multimodality does particularly facilitate affirmative responses— and speaking against the *multimodality for compensation* account. In Experiment 3, we will investigate whether these findings also hold true when processing hand-gestures instead of head-gestures investigated here.

Experiment 3

Transparency and Openness

The experimental setup was identical to Experiment 2, but this time hand gestures were investigated; therefore, to avoid redundancy, only aspects differing from Experiment 2 will be explicitly mentioned in the following section. The experiment was preregistered (https://aspredicted.org/nvzn-hq83.pdf). Data collection was completed between October 11, 2024 and April 16, 2025.

Method

Participants

As in Experiment 2, a necessary sample size of 131 was estimated using the R-package *Superpower*, version 0.2.3 (Lakens & Caldwell, 2021), and the data from Experiment 1. Thus, initially 150 participants were recruited to account for potential exclusions. A procedural error with Prolific participation filters when this experiment was first uploaded led to some non-native German speakers participating in the experiment and subsequently, due to their chance-level performance, a higher rate of participant exclusions resulted for this first batch. This error was corrected for all subsequent participants. Thus, as preregistered, the excluded participants were replaced until a final sample size of 131 was reached. In total, 171 participants ($M_{\text{age}} = 27.62$ years, $SD_{\text{age}} = 9.75$, *range* = 18–62, 94 women, 77 men) took part in the experiment. Out of those, 149 participants stated that they were right-handed (22 left-handed). Due to an overall ER > 30%, a total of 40 participants were excluded from all analyses.

Stimuli and Apparatus

In Experiment 3, only the hand gestures (*thumbs-up* or *thumbs-down*) were investigated. All modifications to create the two unimodal conditions were identical to Experiment 2.

Procedure

The experimental procedure was identical to the one described in Experiment 2. The experimental duration across all participants was approximately 50 minutes.

Design

The experimental design was identical to the one described in Experiment 2.

Results

Analyses were in line with the pre-registration; If there were any departures or additional analyses, these are detailed in the text. In line with Experiment 2—and deviating from the preregistration—all main analyses were conducted using 2x3 repeated measures ANOVAs—again all results relevant for hypothesis testing were not affected by this change. If necessary, Greenhouse-Geisser corrections were applied to account for violations of sphericity.

Main Analysis: Compatibility x Polarity

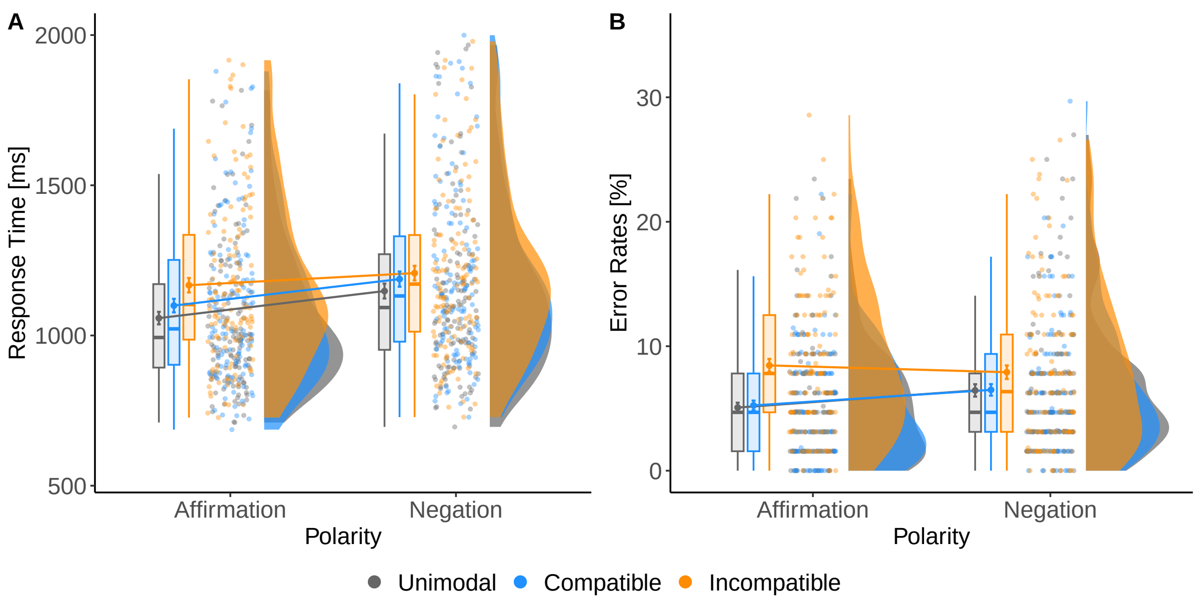
The results are displayed in Figure 4. As in Experiment 2, RTs were shorter for affirmative (M = 1133 ms, SEM = 26) compared to negative trials (M = 1216 ms, SEM = 29), $F(1, 130) = 165.32$, $p < .001$, $\eta_p^2 = 0.56$. There was also a significant effect of compatibility, $F(1.68, 218.57) = 136.02$, $p < .001$, $\eta_p^2 = 0.51$, $\epsilon = 0.84$. Follow-up analyses revealed a significant difference between unimodal (M = 1124 ms, SEM = 26) and compatible (M = 1173 ms, SEM = 28) trials, $t(130) = 7.03$, $p < .001$, $d_z = 0.61$, with unimodal trials again eliciting the fastest responses. There was also a significant difference between incompatible (M = 1226 ms, SEM = 29) and compatible trials, $t(130) = 11.38$, $p < .001$, $d_z = 0.99$. As in Experiment 2, the incompatible trials elicited the slowest responses.

Again, there was a significant interaction between compatibility and polarity on the RTs, $F(1.84, 238.64) = 21.64$, $p < .001$, $\eta_p^2 = 0.14$, $\epsilon = 0.92$. Follow-up analyses revealed that the interaction between compatibility and polarity was also significant when only including compatible and incompatible trials in the analysis, $F(1, 130) = 23.60$, $p < .001$, $\eta_p^2 = 0.15$. There was a larger difference between compatible and incompatible trials for affirmative ($M_D = 74$ ms, $SEM = 7$; $t(130) = 10.88$, $p < .001$, $d_z = 0.95$) compared to negative trials ($M_D = 32$ ms, $SEM = 6$; $t(130) = 5.42$, $p < .001$, $d_z = 0.47$). In line with previous results, the compatibility effect was largest for the affirmative trials— thus not providing any evidence for the *multimodality for compensation* hypothesis. There was no significant interaction between compatibility and polarity between unimodal and compatible trials only, $F(1, 130) = 0.12$, $p = 1.000$, $\eta_p^2 = 0.00$. There was no evidence for a (larger) difference between compatible and unimodal trials for affirmative ($M_D = 50$ ms, $SEM = 7$; $t(130) = 6.81$, $p < .001$, $d_z = 0.59$) compared to negative trials ($M_D = 48$ ms, $SEM = 8$; $t(130) = 5.83$, $p < .001$, $d_z = 0.51$). Again unimodal trials elicited the fastest responses across polarity conditions (see Table C1 and Table C2).

There was a significant effect of polarity on the ERs, $F(1, 130) = 8.61$, $p = .004$, $\eta_p^2 = 0.06$. Affirmative ($M = 6.26$ %, $SEM = 0.37$) trials elicited fewer errors than negative ones ($M = 6.95$ %, $SEM = 0.44$). There was also a significant effect of compatibility on ERs, $F(1.91, 248.26) = 39.40$, $p < .001$, $\eta_p^2 = 0.23$. Follow-up analyses revealed no evidence for a difference between unimodal ($M = 5.76$ %, $SEM = 0.40$) and compatible ($M = 5.87$ %, $SEM = 0.39$) stimuli, $t(130) = 0.38$, $p = 1.000$, $d_z = 0.03$. There was a significant difference between compatible and incompatible stimuli ($M = 8.19$ %, $SEM = 0.48$), with incompatible stimuli producing the largest ERs, $t(130) = 6.86$, $p < .001$, $d_z = 0.60$. There was also a significant interaction of polarity and compatibility on the ERs, $F(1.94, 252.72) = 7.24$, $p < .001$, $\eta_p^2 = 0.05$. Follow-up analyses revealed no evidence for a significant interaction between compatibility and polarity when including compatible and unimodal trials only in the analysis, $F(1, 130) = 0.06$, $p = 1.000$, $\eta_p^2 = 0.00$. Follow-up analyses revealed no evidence for a (larger) difference between compatible and unimodal trials on affirmative (M_D = 0.17 %, *SEM* = 0.38; $t(130) = 0.45$, $p = 1.000$, $d_z = 0.04$) compared to negative trials (M_D = 0.04 %, *SEM* = 0.39; $t(130) = 0.11$, $p = 1.000$, $d_z = 0.01$). There was a significant interaction between compatibility and polarity when including compatible and incompatible trials in the analysis only, $F(1, 130) = 10.32$, $p = .005$, $\eta_p^2 = 0.07$. This was reflected in a larger difference between compatible and incompatible trials on affirmative (M_D = 3.21 %, *SEM* = 0.42; $t(130) = 7.72$, $p < .001$, $d_z = 0.67$) compared to negative trials (M_D = 1.42 %, *SEM* = 0.46; $t(130) = 3.09$, $p = .005$, $d_z = 0.27$). Incompatible trials accounted for the largest ERs across conditions (see Table C1 and Table C3).

Figure 4

A: Reaction Times in milliseconds, B: Error Rates in % for Experiment 3



Note. Within both plots, grouping by Compatibility Condition in Color and Polarity on the X-Axis. Participants individual means with conditions are included within both plots, aggregated means (across participants) are also included within both plots. All aggregated means are surrounded by error bars indicating their respective standard error (+/- 1 SEM).

Additional Analysis: The Influence of Modality

As in Experiment 2, additional preregistered analyses that included the factor modality were conducted for more detailed insights into the influence of participants responding to the verbal or gestural information with regard to the *multimodality for* *compensation* account. All analyses conducted in this subsection were part of a 2x2x3 repeated-measures ANOVA.

Interactions and main effects reported in the main analysis were mirrored by the present results in both RTs and ERs— detailed analysis can be found in Appendix Section C, Table C4 and Table C5. Most interestingly for the hypothesis under investigation, the three-way interaction between compatibility, polarity and modality did not reach significance, $F(1.76, 228.65) = 3.12$, $p = .053$, $\eta_p^2 = 0.02$, $\epsilon = 0.88$. Thus, in line with the findings from Experiment 2, there was no evidence that the main results were significantly affected by whether participants responded to verbal or gestural information.

However, there was again an interesting pattern revealed in this sub-analysis, that mirrors the one reported in Experiment 2. There was an interaction between compatibility and modality, $F(1.61, 208.84) = 60.28$, $p < .001$, $\eta_p^2 = 0.32$, $\epsilon = 0.80$. Looking into this interaction, when responding to verbal information—where the unimodal condition consisted only of speech—then the incompatible condition (M = 1218 ms, SEM = 32) was the slowest, followed by the unimodal condition (M = 1166 ms, SEM = 30), $t(130) = 6.45$, $p < .001$, $d_z = 0.56$. The compatible condition (M = 1144 ms, SEM = 29) elicited the fastest responses, significantly faster than the unimodal condition, $t(130) = 3.81$, $p < .001$, $d_z = 0.33$. This was not the case when gesture was the response-relevant dimension—and thus the unimodal condition consisted of only gestures. In this case, the incompatible condition (M = 1239 ms, SEM = 30) was the slowest again, this time, followed by the compatible condition (M = 1205 ms, SEM = 30), $t(130) = 5.64$, $p < .001$, $d_z = 0.49$. The unimodal condition (M = 1083 ms, SEM = 26) elicited the fastest responses, $t(130) = 9.59$, $p < .001$, $d_z = 0.84$.[8]

Within the ERs, the critical three-way interaction between compatibility, modality and polarity did not reach significance ($F(1.96, 254.27) = 4.20$, $p = .017$, $\eta_p^2 = 0.03$, ϵ

[8] It may be relevant to mention, again, identical to Experiment 2, despite the three-way interaction not reaching significance, compatible trials only elicited faster responses than unimodal trials in the affirmative condition $t(130) = 4.70$, $p < .001$, $d_z = 0.41$. There was no evidence for a difference between compatible and unimodal trials in the negative condition, $t(130) = 1.66$, $p = .300$, $d_z = 0.14$. Thus, this result pattern is in line with the idea that compatibility is specifically helpful in affirmative compared to negated trials.

= 0.98)—thus, there was no evidence that modality had a significant influence on the critical interaction.

Discussion

This study replicated the central result patterns found in Experiments 1 and 2, this time investigating hand gestures only. The central findings were processing difficulties both for incompatible and negated stimuli. Additionally, participants seemed to particularly benefit from the presence of multimodality in the affirmative condition again — a finding not supporting a *multimodality for compensation* account. As in Experiment 2, participants behaved very similarly in the unimodal condition—with this being the fastest if the relevant response dimension was gesture-only. In contrast, when the relevant response dimension was verbal, follow-up tests showed that compatible condition was the fastest. The implications of these findings will be discussed in the following sections.

General Discussion

The present study investigated the processing of multimodal information when comprehending affirmative or negative answers to polar questions. Previous research has shown that multimodal information, in the form of iconic gestures, speeds up the processing of questions (see Ter Bekke et al., 2024a), facilitates comprehension in ambiguous contexts (Holle & Gunter, 2007), and aids comprehension in difficult-to-process situations (Billot-Vasquez et al., 2020; Krason et al., 2022). And more generally, it has been shown that multimodal information can be used to predict upcoming information and thus ease comprehension (see Ter Bekke et al., 2025). While research has often focused on the processing of iconic gestures, comparatively little attention has been given to emblematic gestures, whose meanings are more symbolic in nature and culturally agreed upon (see Andric et al., 2013; Gawne & Cooperrider, 2024). Thus, here we investigated the integration of emblematic gestures when processing yes-no answers to polar questions. Interestingly, it has become somewhat of a consensus that processing negation encompasses processing difficulties across a wide range of phenomena (for an overview, see Kaup & Dudschig, 2020; Wales & Grieve, 1969). Thus, as negated statements are typically more cognitively demanding to process, we aimed to test the *multimodality for compensation* account within this context of negation. In order to do so, we adapted a basic paradigm that was introduced to study negation comprehension in children (Austin et al., 2014; Feiman et al., 2017) to an adult setup that allowed for measuring processing speed. Specifically, following a question regarding the unknown location of a target item (i.e. a ball), participants viewed videos that provided them with the correct answer for locating the target item. The answer could either be affirmative or negative ("Is the ball in the blue box?" Video: "yes/no" verbal and/or gestural). Across three experiments and two gesture types (head and hand gestures), the central findings were replicated. These will be discussed in detail in the following sections.

Negation

Negation processing is a central aspect of everyday language, and every language has means of expressing negation. Nevertheless, the literature has consistently shown that negation often poses challenges on processing (see Hasson & Glucksberg, 2006; Kaup & Dudschig, 2020; Wales & Grieve, 1969). In line with this hypothesis, the present findings reflect the posited adverse impact of negation. Participants performances were significantly exacerbated by negation—both when responses were made to gestural or verbal information. This effect held true even in the unimodal condition, and despite participants having plenty of time to prepare for the subsequent response cue. One might assume that participants potentially had the option to build an alternative representation (i.e. "if he/she says yes, I press the right button"), which might potentially overcome the processing difficulties associated with negation. The finding that participants cannot use this extended preparatory time to reduce the impact of negation is indeed in line with the literature on negation processing in basic instructions (see Dudschig et al., 2019) and sentences (see Lago et al., 2025). Importantly, the present paradigm was adapted from previous experimental studies investigating the age of acquisition of negation (Austin et al., 2014; Feiman et al., 2017). This setup was chosen as both affirmative and negated answers should be similarly pragmatically felicitous responses to the question posed (see Espinal & Tubau, 2019; Krifka, 2013; Roelofsen & Farkas, 2015; Yadugiri, 1986). Thus, potential negation processing difficulties are likely due to inherent processing difficulties associated with negation, rather than missing licensing contexts (see Nieuwland & Kuperberg, 2008; Orenes et al., 2016)—at least when considering response particles as in the current study. Overall, these findings neatly fit in with and extend previous research on the adverse effects of processing negated statements (Dudschig et al., 2021) to the processing of gestural and multimodal negated information.

Multimodality

The present study was not only concerned with the processing of negation across modalities but, more centrally, with the integration of multimodal input during comprehension —specifically the question whether multimodality is especially helpful in cognitively demanding situations—in this case negation. Across three experiments, it was shown that multimodal information becomes directly integrated during processing—even when the information was fully task-irrelevant. This integration was reflected in processing benefits for compatible compared to incompatible conditions—both when responding to verbal or to gestural information. These findings further support the notion of a unified system for gestural and verbal integration during the comprehension process (see Holler & Levinson, 2019; Kelly, Creigh, & Bartolotti, 2010; Kendon, 2000; Kita & Emmorey, 2023; Vainiger et al., 2014).

Prior research has demonstrated that multimodal gestural input can be especially advantageous under conditions of heightened comprehension difficulty (see Billot-Vasquez et al., 2020; Drijvers & Özyürek, 2020; Kartalkanat & Göksun, 2020; Krason et al., 2022). Accordingly, it was first hypothesized that, due to the increased cognitive demands associated with negation processing, the facilitative effects of multimodal information would be more pronounced for negated statements than for affirmative ones —summarized by the *multimodality for compensation* account. However, Experiment 1 showed that across both gesture types (head gestures and hand gestures), multimodality facilitated the processing of compatible negated responses—but even more so, the processing of compatible affirmative responses. This finding was further supported in Experiments 2 and 3, suggesting that it is robust. So why might this be the case? At present, the issue remains open to debate, and we can only hypothesize about the potential reasons. We address one of these potential reasons below.

One might speculate that negation gestures are less frequent or less natural during processing speech than affirmative gestures. However, as mentioned earlier, the variety of negation gestures is higher than for affirmation gestures (Gawne, 2021; Harrison, 2024). With regard to frequency, the findings in the literature lack consensus. For example, some studies suggest that negation gestures—specifically head shaking—emerge earlier in child development than affirmative gestures like head nodding (see Fenson et al., 1994; Kettner & Carpendale, 2013). Interestingly, when looking at mothers' communicative input in mother-child interaction, they seem to produce more affirmative gestures such as head nods in relation to polar yes/no questions—contrasting the children production data (see Fusaro et al., 2014). Children at the age of as early as 12 months already produce head shakes, and with 14 months, head nods, according to a large study using parents reports in the U.S. (see Fenson et al., 1994). Also, a study in French showed that children at the age of 16 months predominantly communicate refusal or agreement with gestures, but in the following year use all types of formats, verbal-only, speech-gesture combinations and gesture-only (see Guidetti, 2005). Some studies even suggest that the head-shake is evolutionarily rooted in children refusing food intake—and thus has a straight trajectory evolving into a sign of negation later in life (see Bross, 2020). Adult data on the production and thus the frequency of negative and affirmative gestures is rare (Kelmaganbetova et al., 2023; Loos & Repp, in press), but there is consensus that all the investigated gestures are standard, well-known gestures in adult language, both in the affirmative and negated case (Austin et al., 2014; Ekman & Friesen, 1969; Fusaro et al., 2012, 2014; Kelmaganbetova et al., 2023; McNeill, 1992). Similarly, for the hand gestures examined in this study, there is limited research on their age of acquisition or production frequency. However, it is well established that both the affirmative and negated versions are widespread conversational gestures in Western societies (Gawne & Cooperrider, 2024). Taken together, there is currently no strong reason to assume that the negation gestures implemented in this study are less frequently occurring or less familiar to adults, making frequency an unlikely explanation for their reduced effectiveness. Instead, the findings suggest that while compatible multimodal cues can ease negation processing, they do so less effectively than for affirmatives.

In summary, the observation that specifically affirmation—more than negation—benefits from simultaneous gestural input remains surprising in light of the extensive evidence demonstrating that gestures generally aid comprehension under conditions of increased processing difficulty (e.g., degraded auditory input, ambiguity, or hearing loss). Importantly, previous studies have rarely compared linguistic stimuli differing in complexity or processing demands when addressing this issue; rather, they have tended to examine whether gesture use increases under conditions of imposed difficulty. Also interestingly, previous research has indicated that negation is special, not only due to its increased processing demands, but also due to potentially additional processing steps during integration (Dudschig & Kaup, 2018) and due to its specific pragmatic requirements (Givón, 1978). Thus, negation might be “special” because it challenges the language system across multiple levels — cognitive (maintaining / rejecting alternatives), pragmatic (contextual expectations), and neural (increased control and integration demands). As a result, one possible explanation is that negation represents a unique linguistic operator; consequently, the *multimodality for compensation* hypothesis warrants further examination across a broader range of linguistically complex phenomena.

Another key question in this study was how multimodal information, whether compatible or incompatible, compares to unimodal processing. Across two experiments and both gesture types (head gestures and hand gestures), unimodal processing was consistently faster when looking at the overall result pattern — even when compared to compatible multimodal input (with only one notable exception). Thus, while initially counterintuitive, this finding warrants further consideration of potential underlying causes, which might be of interest for future investigations and a better overall picture of gesture integration. Detailed follow-up analyses distinguishing conditions in which participants responded to verbal versus gestural input (Additional Analysis: The Influence of Modality, Experiments 2 and 3) indicated that, when responding to verbal information—and thus the unimodal signal being the speech-only signal—participants were fastest in the multimodal compatible condition for affirmative trials. For negated trials, however, response times did not differ significantly between the unimodal and multimodal compatible conditions. This effect was observed consistently across both head (Experiment 2) and hand (Experiment 3) gestures. These results imply that multimodal cues can enhance processing efficiency during emblem processing, but primarily when the task demands focus on less demanding (i.e. affirmative) verbal input. Arguably looking at responses to verbal information only, resembles the most ecologically valid communication scenario—however, importantly even in this condition negation benefited less from multimodality than affirmation—again speaking against the idea of a *multimodality for compensation* account.

Previous studies have often shown that gestures accompanying speech generally benefit comprehension (Hostetter, 2011). This is somehow different from the present study, where a differential impact of affirmative to negated gestures was found. However, these previous studies varied in some key aspects, which will be discussed in the following sections. First, one major difference in previous studies was that often iconic gestures were used to investigate gesture integration. For example, in a very recent study conducted by Ter Bekke et al. (2024a), it was shown that iconic gestures produced during the question phase (e.g. "Is being able to type well useful when you are a secretary?" accompanied by a typing gesture) results in faster responses compared to conditions without gestures (see also Holler et al., 2018). Iconic gestures typically visually represent or mimic the event, entity or an attribute of an entity under discussion (e.g., showing swimming movements when gesturing about a swimming event; gesturing a round shape when talking about a football). Accordingly, such gestures may contribute to enhanced processing rather generally, irrespective of the specific gesture. In contrast, the differential effect of different types of gestures in the present study, might be due to the use of emblematic gestures. Emblems, in the context of conversation, have an established and agreed upon meaning in a specific group and thus are culturally acquired (Kendon, 1981; McNeill, 1992). It is, however, not always definite which specific category a certain gesture belongs to, and the difference between the types of gestures is less distinctive than often assumed (see Kendon, 2002). Nevertheless, both types of gestures investigated here can be summarized under the category of emblems, which are also defined by being "those nonverbal acts which have a direct verbal translation" (see Ekman & Friesen, 1969, p. 63). Thus, it is possible that the speech-gesture combinations, when using emblems, are less beneficial than combining speech with iconic gestures, which also fits with meta-analytical results suggesting that the influence of gestures reduces with increasing redundancy between gesture and speech (see Hostetter, 2011). In fact, some authors have argued that emblems resemble speech more closely than other types of gestures, such as representational or metaphoric gestures (see, Kita & Emmorey, 2023). This might be a factor that contributes to an increasing redundancy and thereby hinder, instead of facilitating, processing—for example by the need to access similar processing resources. However, it would still remain unclear why this should be specifically the case for negated answers—where redundant gestures help even less than in affirmative cases. To further investigate this issue, a direct comparison between the benefits of more diverse emblematic gestures and with iconic gesture types may prove particularly insightful. Additionally, it is possible—also speculative at the moment—that integration is primarily driven by movement similarity between speech and gesture, which has been shown to modulate the magnitude of imitation effects (Leighton & Heyes, 2010). Consequently, these effects may be more pronounced for iconic gestures, as the degree of movement overlap (e.g. gesturing "round" and saying "round") can be greater than for emblematic gestures[9].

To our knowledge, this study is among the first to point to such differences in processing between certain gesture types (see Chui et al., 2018). One other reason

[9] We thank a reviewer for bringing this to our attention.

might be that while emblems are more abstract, iconic gestures serve as visual representations of the described events or referents and thus help create the embodied representations sometimes assumed to lie at the core of meaning comprehension (see Barsalou, 2008; Dudschig et al., 2016; Glenberg & Kaschak, 2002; Schütt et al., 2023; Winter et al., 2022). Therefore, the effectiveness of emblematic gestures in supporting processing may be influenced by how abstract the specific gesture is — a factor that may be particularly pronounced for negation (see Pea, 1978).

On the other hand, it is also important to note that some of the current findings may also stem from specific design choices within the paradigm itself. While prior research on gesture processing has examined differences between unimodal and multimodal conditions, there is still no clear consensus on how best to approach this question. Many studies address the broader issue of whether gestures are spontaneously integrated, a question that does not necessarily require a unimodal condition. For instance, some studies have compared the effects of compatible multimodal input to pairings of speech with nonsensical gestures (Chui et al., 2018; Gunter & Bach, 2004; Holle & Gunter, 2007), while others have focused on contrasts between compatible and incompatible gesture-speech combinations (Kelly, Özyürek, & Maris, 2010; Kelly et al., 2007; Noah et al., 2017). Also in some studies, the “neutral” condition sometimes consisted of random gesticulation that held no relevance to the utterance— such as actors scratching their chin (e.g., Holle & Gunter, 2007)—a design choice that might be more relevant than sometimes assumed. In some cases, a unimodal condition was included and compared directly to compatible multimodal input (Drijvers & Holler, 2023; Krason et al., 2022). Thus, the way compatible and incompatible information is operationalized and intermixed varies substantially across studies, which may strongly influence the observed processing patterns. Therefore, to allow for definitive conclusions regarding potential integration differences of iconic and emblematic gestures, future research should directly compare these factors within the same experimental framework.

Related to this issue is the choice of the proportions of multimodal, unimodal and compatible or incompatible information. In the present study, each condition (multimodal-compatible, multimodal-incompatible, unimodal) was presented in one third of the trials. Thus, participants might realize that speech accompanying gestures are only helpful in half of the multimodal trials (the compatible ones) and hinder processing in the other half of the multimodal trials (the incompatible ones). Thus, participants might be able to reduce the influence of the irrelevant task dimension—resulting in a generally reduced influence of compatibility. Indeed, this is in line with an example of an attenuated effect of multimodality in the literature put forward by Holle and Gunter (2007), who found a weakened influence of gestures when they added uninformative gestures to the mix, thereby reducing the informational value of multimodality.

Interestingly, there are also findings in the basic conflict processing literature (Bodner & Dypvik, 2005; Eriksen & Eriksen, 1974; Kinoshita et al., 2011; Smith & Ulrich, 2024) that, while somewhat peripheral to the primary focus of the present research, nevertheless remain both relevant and noteworthy. In the Flanker task—a basic conflict processing task—a central letter serves as target cue and it is flanked by compatible or incompatible task-irrelevant stimuli (e.g., incompatible: « > «; compatible: » > »). Already in the original study (Eriksen & Eriksen, 1974) faster response times were observed in the absence of irrelevant information—regardless of whether that information might be compatible (see also Smith & Ulrich, 2024). In other words, in this study, participants were fastest if only the central target cue was presented—even in comparison to the compatible condition. Comparisons with unimodal conditions in the conflict literature suggest that, while such comparisons are central, the question of how to appropriately design a unimodal condition remains a topic of debate even after decades of research. Similarly, the present study demonstrated that the choice of the relevant unimodal condition—gesture versus speech—can substantially influence the observed effects. Although this comparison between general conflict processing and conflict processing in multimodal language input may initially appear speculative, an increasing body of literature indicates that language processing and general conflict processing—even in basic tasks—share notable similarities and are governed by comparable underlying mechanisms (Dudschig, 2022; Dudschig & Kaup, 2018). Thus, it likely would be beneficial for future research to closer investigate whether mechanisms from conflict processing also apply to the processing of task-irrelevant gestures and speech situations. One example in standard conflict literature would be the *proportion congruency effect* (Bodner & Dypvik, 2005; Kinoshita et al., 2011), where it is routinely observed that the congruency effect is attenuated under more demanding circumstances (a higher proportion of incongruent trials), supposedly because cognitive control is exerted to a higher degree. In the present example, the rather high proportion of incompatible language information might lead to a more top-down processing style (i.e. a higher exertion of cognitive control) and reduced influence of a task-irrelevant dimension. Bringing together these research fields might be a promising avenue for future studies.

Conclusion

The present study examined the integration of emblematic gestures during language comprehension. The findings demonstrate that multimodal speech-gesture information is indeed directly integrated during processing, even when it involves emblems and when information is fully task-irrelevant. Moreover, the study reveals that processing difficulties associated with negation persist in multimodal contexts. While compatible multimodal input generally facilitates comprehension, this facilitation is significantly stronger for affirmative statements compared to negative ones. Thus, across all studies, we found no evidence supporting the *multimodality-for-compensation* hypothesis. This suggests that gestures do not provide particular benefits for comprehenders when processing linguistically demanding information—specifically, negation in our case. These findings shed light on potential limits of multimodality in language processing and point towards the importance of future studies investigating the role of specific gesture types. Future research could also explore the extent to which these effects of integrating emblematic gestures parallel standard conflict processing mechanisms, potentially offering new insights into the cognitive integration of gestures in language understanding.

Constraints on Generality

With regard to generality across different participant samples, it is important to note that online data collection via Prolific should reduce the likelihood of only testing a very specific sample of psychology students in the first years of study (Gosling et al., 2010)—which is already indicated by the comparably high ratio of male participants and wider age range. Also, with regard to generality across different languages we currently have no reason to assume that the processing of affirmative and negated answers in the multimodal context would differ. However, this remains an empirical question that offers a potential avenue for future research. Importantly, we think that our findings might indeed be specific to the use of emblematic gestures and basic yes/no answers—thus, the *multimodality-for-compensation* hypothesis might receive support with other linguistic constructions, or in situations where comprehension difficulties arise not from linguistic complexity itself but from external factors (e.g., background noise, Krason et al., 2022). However, we also show generality across two types of emblematic gestures (head- and hand gesture) and replicate the central findings across three experiments, thus we assume rather high generality with regard to the specific form of negation/affirmation gesture. It must also be noted that the current experiment involved the presentation and interpretation of multimodal input via the playback of pre-recorded videos. It is likely that face-to-face two-way communication involves more complex nuances that are difficult to investigate within a controlled experimental setup, thus making it important that future studies extend the limited set of stimuli used in the present work. However, practical difficulties aside, this might also offer a promising avenue for the investigation of affirmative and negated gestures within an even more naturalistic setting.

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Appendix A

Experiment 1

Table A1

Aggregated RTs and ERs across Polarity × Compatibility × Modality × Gesture for Experiment 1

Polarity	Compatibility	Modality	Gesture	RT	ER
Affirmation	Compatible	Gesture	Head	1,018 (17)	3.72 (0.41)
Affirmation	Compatible	Gesture	Thumb	1,052 (17)	4.64 (0.45)
Affirmation	Compatible	Speech	Head	976 (18)	3.69 (0.41)
Affirmation	Compatible	Speech	Thumb	995 (17)	3.85 (0.45)
Affirmation	Incompatible	Gesture	Head	1,132 (20)	7.87 (0.70)
Affirmation	Incompatible	Gesture	Thumb	1,105 (19)	7.61 (0.64)
Affirmation	Incompatible	Speech	Head	1,111 (21)	8.82 (0.70)
Affirmation	Incompatible	Speech	Thumb	1,057 (20)	5.95 (0.57)
Negation	Compatible	Gesture	Head	1,171 (21)	5.48 (0.49)
Negation	Compatible	Gesture	Thumb	1,166 (20)	6.95 (0.64)
Negation	Compatible	Speech	Head	1,096 (22)	5.53 (0.53)
Negation	Compatible	Speech	Thumb	1,092 (20)	4.98 (0.48)
Negation	Incompatible	Gesture	Head	1,171 (20)	5.88 (0.48)
Negation	Incompatible	Gesture	Thumb	1,173 (19)	7.60 (0.59)
Negation	Incompatible	Speech	Head	1,140 (24)	8.25 (0.73)
Negation	Incompatible	Speech	Thumb	1,123 (23)	6.79 (0.63)

Note. RTs in milliseconds, ERs in %. *SEM*s of all aggregated values are included in parentheses.

Table A2

Statistical Results, for the Main Analyses in Experiment 1.

Reaction Times			
Effect	Test Statistic (F/t)	p	$p < .05$
Compatibility	$F(1, 145) = 275.44$	$< .001$	***
Polarity	$F(1, 145) = 244.16$	$< .001$	***
Compatibility:Polarity	$F(1, 145) = 131.41$	$< .001$	***
Follow-up analysis, Compatibility grouped by Polarity:			
Aff	$t(145) = 17.97$	$< .001$	***
Neg	$t(145) = 5.04$	$< .001$	***
There was a significant effect of both Compatibility and Polarity on RTs. There was also an interaction between both factors that was due to a larger effect of compatibility on affirmative compared to negative trials.			
Error Rates			
Effect	Test Statistic (F/t)	p	$p < .05$
Compatibility	$F(1, 145) = 99.52$	$< .001$	***
Polarity	$F(1, 145) = 5.88$	$.017$	*
Compatibility:Polarity	$F(1, 145) = 22.96$	$< .001$	***
Follow-up analysis, Compatibility grouped by Polarity:			
Aff	$t(145) = 9.75$	$< .001$	***
Neg	$t(145) = 4.54$	$< .001$	***
There was a significant effect of both Compatibility and Polarity on ERs. There was also an interaction between both factors that was due to a larger effect of compatibility on affirmative compared to negative trials.			

Note. * < .05, ** < .01, *** < .001, Aff stands for affirmative and Neg for negative. The factor Compatibility consists of the levels compatible and incompatible, the factor Polarity is made-up of affirmative and negative trials.

Table A3
Statistical Results, Additional analysis on the Influence of Modality in Experiment 1 (Part I).

Reaction Times			
Effect	Test Statistic (F/t)	p	$p < .05$
Modality	$F(1, 145) = 11.11$	.001	**
Compatibility	$F(1, 145) = 269.11$	< .001	***
Polarity	$F(1, 145) = 228.91$	< .001	***
Compatibility:Polarity	$F(1, 145) = 127.54$	< .001	***
Follow-up analysis, Compatibility grouped by Polarity:			
Aff	$t(145) = 18.02$	< .001	***
Neg	$t(145) = 4.91$	< .001	***
There was a significant interaction of Compatibility and Polarity on RTs. This was again due to a larger effect of compatibility on affirmative compared to negative trials.			
Compatibility:Modality	$F(1, 145) = 14.65$	< .001	***
Follow-up analysis, Compatibility grouped by Modality:			
Gesture	$t(145) = 9.54$	< .001	***
Speech	$t(145) = 14.34$	< .001	***
There was a significant interaction of Compatibility and Modality on RTs. This was due to a larger effect of compatibility on verbal compared to gestural trials.			

Note. * < .05, ** < .01, *** < .001, Aff stands for affirmative and Neg for negative. The factor Compatibility consists of the levels compatible and incompatible, the factor Polarity is made-up of affirmative and negative trials. The factor Modality encompasses gestural and verbal trials and describes which dimension was response-relevant.

Table A4

Statistical Results, Additional analysis on the Influence of Modality in Experiment 1 (Part II).

Reaction Times			
Effect	Test Statistic (F/t)	p	$p < .05$
Polarity:Modality	$F(1, 145) = 4.27$	.041	*
Follow-up analysis, Polarity grouped by Modality:			
Gesture	$t(145) = 14.05$	< .001	***
Speech	$t(145) = 11.16$	< .001	***
There was a significant interaction of Polarity and Modality on RTs. This was due to a larger effect of Polarity on gestural compared to verbal trials.			
Compatibility:Polarity:Modality	$F(1, 145) = 3.16$	.077	
Follow-up analysis, Compatibility:Polarity grouped by Modality:			
Gesture	$F(1, 145) = 81.80$	< .001	***
Speech	$F(1, 145) = 58.04$	< .001	***
The three-way interaction between Compatibility, Polarity and Modality did not reach significance.			

Note. * < .05, ** < .01, *** < .001, Aff stands for affirmative and Neg for negative. The factor Compatibility consists of the levels compatible and incompatible, the factor Polarity is made-up of affirmative and negative trials. The factor Modality encompasses gestural and verbal trials and describes which dimension was response-relevant.

Table A5

Statistical Results, Additional analysis on the Influence of Gesture Type in Experiment 1 (Part I).

	Reaction Times		
Effect	Test Statistic (*F*/*t*)	*p*	*p* < .05
Gesture Type	$F(1, 145) = 4.73$	.031	*
Compatibility	$F(1, 145) = 274.57$	< .001	***
Polarity	$F(1, 145) = 243.45$	< .001	***
Compatibility:Polarity	$F(1, 145) = 131.92$	< .001	***
Follow-up analysis, Compatibility grouped by Polarity:			
Aff	$t(145) = 17.92$	< .001	***
Neg	$t(145) = 4.99$	< .001	***
There was a significant interaction of Compatibility and Polarity on RTs. This was again due to a larger effect of compatibility on affirmative compared to negative trials.			
Compatibility:Gesture	$F(1, 145) = 34.74$	< .001	***
Follow-up analysis, Compatibility grouped by Gesture:			
Head	$t(145) = 14.98$	< .001	***
Hand	$t(145) = 9.34$	< .001	***
There was a significant interaction of Compatibility and Gesture Type on RTs. This was due to a larger effect of compatibility on head compared to hand gestures.			

Note. * < .05, ** < .01, *** < .001, Aff stands for affirmative and Neg for negative. The factor Compatibility consists of the levels compatible and incompatible, the factor Polarity is made-up of affirmative and negative trials. The factor Gesture encompasses hand and thumb gestures and describes the kind of the produced gestures (thumbs up/down vs. head shake/nod).

Table A6

Statistical Results, Additional analysis on the Influence of Gesture Type in Experiment 1 (Part II).

Reaction Times			
Effect	Test Statistic (*F*/*t*)	*p*	*p* < .05
Polarity:Gesture	$F(1, 145) = 0.03$	.868	
Follow-up analysis, Polarity grouped by Gesture:			
Head	$t(145) = 13.19$	< .001	***
Hand	$t(145) = 15.39$	< .001	***
There was no significant interaction between Polarity and Gesture Type. Thus, there was no evidence that the gesture type influences the effect of Polarity.			
Compatibility:Polarity:Gesture	$F(1, 145) = 29.50$	< .001	
Follow-up analysis, Compatibility:Polarity grouped by Gesture:			
Head	$F(1, 145) = 112.11$	< .001	***
Hand	$F(1, 145) = 30.45$	< .001	***
The three-way interaction between Compatibility, Polarity and Gesture Type did reach significance. Thus indicating, as seen in the Follow-up analyses, that the crucial interaction between Compatibility and Polarity was modulated by the Gesture.			

Note. * < .05, ** < .01, *** < .001, Aff stands for affirmative and Neg for negative. The factor Compatibility consists of the levels compatible and incompatible, the factor Polarity is made-up of affirmative and negative trials. The factor Gesture encompasses hand and thumb gestures and describes the kind of the produced gestures (thumbs up/down vs. head shake/nod).

Appendix B

Experiment 2

Table B1

Aggregated RTs and ERs across Polarity × Compatibility × Modality for Experiment 2

Polarity	Compatibility	Modality	RT	ER
Affirmation	Compatible	Gesture	1,025 (27)	4.25 (0.44)
Affirmation	Compatible	Speech	1,034 (23)	4.18 (0.41)
Affirmation	Incompatible	Gesture	1,138 (31)	7.23 (0.66)
Affirmation	Incompatible	Speech	1,180 (26)	10.90 (0.75)
Affirmation	Unimodal	Gesture	934 (25)	4.26 (0.47)
Affirmation	Unimodal	Speech	1,091 (24)	4.45 (0.52)
Negation	Compatible	Gesture	1,171 (31)	5.95 (0.47)
Negation	Compatible	Speech	1,167 (26)	5.79 (0.49)
Negation	Incompatible	Gesture	1,204 (33)	6.32 (0.55)
Negation	Incompatible	Speech	1,236 (29)	8.59 (0.68)
Negation	Unimodal	Gesture	1,053 (28)	6.24 (0.53)
Negation	Unimodal	Speech	1,174 (25)	4.99 (0.46)

Note. RTs in milliseconds, ERs in %. *SEMs* of all aggregated values are included in parentheses.

Table B2

Statistical Results, for the Main RT Analysis in Experiment 2.

Reaction Times			
Effect	Test Statistic (F/t)	p	$p < .05$
Polarity	$F(1, 137) = 242.34$	< .001	***
Compatibility	$F(1.80, 247.00) = 165.74$	< .001	***
Follow-up analysis, comparing Compatibility levels:			
Comp—Uni	$t(137) = 5.19$	< .001	***
Incomp—Uni	$t(137) = 15.61$	< .001	***
Incomp—Comp	$t(137) = 14.82$	< .001	***
There was a significant effect of Compatibility on RTs. Follow-up analyses revealed that all 3 factor levels differed significantly with Unimodal trials eliciting the fastest responses followed by compatible and lastly incompatible trials.			
Compatibility:Polarity	$F(1.97, 269.21) = 49.78$	< .001	***
Follow-up analysis, Polarity effect across Compatibility levels:			
(Comp, Uni):(Aff, Neg)	$F(1, 137) = 24.88$	< .001	***
(Incomp, Uni):(Aff, Neg)	$F(1, 137) = 27.85$	< .001	***
(Incomp, Comp):(Aff, Neg)	$F(1, 137) = 89.25$	< .001	***
There was a significant interaction between Compatibility and Polarity. The Follow-up analysis indicates that this interaction was present across all three factor levels of compatibility.			

Note. * < .05, ** < .01, *** < .001, Comp stands for compatible, Incomp for incompatible and Uni for unimodal, Aff stands for affirmative, Neg for negative. The Follow-up analysis of the Compatibility:Polarity consist of three different versions of the interaction, within the parentheses, the (abbreviated) levels of each factor for the investigated interaction are listed.

Table B3

Statistical Results, for the Main ER Analysis in Experiment 2.

	Error Rates		
Effect	Test Statistic (F/t)	p	$p < .05$
Polarity	$F(1, 137) = 2.41$	.123	
Compatibility	$F(1.70, 232.59) = 90.41$	< .001	***
Follow-up analysis, comparing Compatibility levels:			
Comp—Uni	$t(137) = 0.27$	1.000	
Incomp—Uni	$t(137) = 10.55$	< .001	***
Incomp—Comp	$t(137) = 10.62$	< .001	***
There was a significant effect of Compatibility on ERs. Follow-up analyses revealed no evidence that unimodal and compatible trials differed. There was however evidence that the incompatible condition differed significantly from both unimodal and compatible trials.			
Compatibility:Polarity	$F(1.84, 252.56) = 23.13$	< .001	***
Follow-up analysis, Polarity effect across Compatibility levels:			
(Comp, Uni):(Aff, Neg)	$F(1, 137) = 0.77$	1.000	
(Incomp, Uni):(Aff, Neg)	$F(1, 137) = 26.77$	< .001	***
(Incomp, Comp):(Aff, Neg)	$F(1, 137) = 33.24$	< .001	***
There was a significant interaction of compatibility and polarity on ERs. Follow-up analyses revealed that this effect was driven by the incompatible trials. There was no evidence for an interaction across compatible and unimodal trials only.			

Note. * < .05, ** < .01, *** < .001, Comp stands for compatible, Incomp for incompatible and Uni for unimodal, Aff stands for affirmative, Neg for negative. The Follow-up analysis of the Compatibility:Polarity consist of three different versions of the interaction, within the parentheses, the (abbreviated) levels of each factor for the investigated interaction are listed.

Table B4

Statistical Results, Additional analysis on the Influence of Modality in Experiment 2 (Part I).

Reaction Times			
Effect	Test Statistic (F/t)	p	$p < .05$
Modality	$F(1, 137) = 11.91$	< .001	***
Polarity	$F(1, 137) = 235.75$	< .001	***
Compatibility	$F(1.82, 248.66) = 173.05$	< .001	***
Follow-up analysis, comparing Compatibility levels:			
Comp—Uni	$t(137) = 5.27$	< .001	***
Incomp—Uni	$t(137) = 15.92$	< .001	***
Incomp—Comp	$t(137) = 14.99$	< .001	***
There was a significant effect of Compatibility on RTs. Follow-up analyses revealed that all 3 factor levels differed significantly with Unimodal trials eliciting the fastest responses followed by compatible and lastly incompatible trials.			
Compatibility:Polarity	$F(1.95, 266.69) = 49.07$	< .001	***
Follow-up analysis, Polarity effect across Compatibility levels:			
(Comp, Uni):(Aff, Neg)	$F(1, 137) = 27.85$	< .001	***
(Incomp, Uni):(Aff, Neg)	$F(1, 137) = 24.92$	< .001	***
(Incomp, Comp):(Aff, Neg)	$F(1, 137) = 86.85$	< .001	***
There was a significant interaction of Compatibility and Polarity on RTs. Follow-up analyses revealed that this interaction was present across all three factor levels of compatibility.			

Note. * < .05, ** < .01, *** < .001, Comp stands for compatible, Incomp for incompatible and Uni for unimodal, Aff stands for affirmative, Neg for negative and Gest stands for gesture. The Follow-up analysis of the Compatibility:Polarity consist of three different versions of the interaction, within the parentheses, the (abbreviated) levels of each factor for the investigated interaction are listed.

Table B5

Statistical Results, Additional analysis on the Influence of Modality in Experiment 2 (Part II).

Reaction Times			
Effect	Test Statistic (F/t)	p	$p < .05$
Compatibility:Modality	$F(1.77, 242.29) = 67.83$	< .001	***
Follow-up analysis, Modality effect across Compatibility levels:			
(Comp, Uni):(Gest, Speech)	$F(1, 137) = 119.63$	< .001	***
(Incomp, Uni):(Gest, Speech)	$F(1, 137) = 54.27$	< .001	***
(Incomp, Comp):(Gest, Speech)	$F(1, 137) = 12.25$	.002	**
There was a significant interaction of Compatibility and Modality on RTs. Follow-up analyses revealed that this interaction was present across all three factor levels of compatibility.			
Polarity:Modality	$F(1, 137) = 6.38$	.013	*
Follow-up analysis, Polarity grouped by Modality:			
Gesture	$t(137) = 13.76$	< .001	***
Speech	$t(137) = 12.62$	< .001	***
There was a significant interaction of Polarity and Modality on RTs. Follow-up analyses revealed a larger effect of Polarity on gestural compared to verbal trials.			
Compatibility:Polarity:Modality	$F(1.73, 236.44) = 1.62$	.202	
Follow-up analysis, Compatibility:Polarity Interaction across Modality levels:			
Gesture (Comp:Pola)	$F(2, 137) = 21.88$	< .001	***
Speech (Comp:Pola)	$F(2, 137) = 23.82$	< .001	***
There was no evidence for a significant three-way interaction between Compatibility, Polarity and Modality.			

Note. * < .05, ** < .01, *** < .001, Comp stands for compatible, Incomp for incompatible and Uni for unimodal, Aff stands for affirmative, Neg for negative and Gest stands for gesture. The Follow-up analysis of the Compatibility:Polarity consist of three different versions of the interaction, within the parentheses, the (abbreviated) levels of each factor for the investigated interaction are listed.

Appendix C

Experiment 3

Table C1

Aggregated RTs and ERs across Polarity × Compatibility × Modality for Experiment 3

Polarity	Compatibility	Modality	RT	ER
Affirmation	Compatible	Gesture	1,155 (29)	5.54 (0.45)
Affirmation	Compatible	Speech	1,097 (28)	4.95 (0.56)
Affirmation	Incompatible	Gesture	1,209 (30)	8.13 (0.54)
Affirmation	Incompatible	Speech	1,195 (32)	8.79 (0.73)
Affirmation	Unimodal	Gesture	1,023 (24)	5.20 (0.41)
Affirmation	Unimodal	Speech	1,128 (28)	4.96 (0.54)
Negation	Compatible	Gesture	1,254 (32)	7.09 (0.56)
Negation	Compatible	Speech	1,190 (31)	5.90 (0.61)
Negation	Incompatible	Gesture	1,270 (31)	6.86 (0.53)
Negation	Incompatible	Speech	1,240 (34)	8.97 (0.77)
Negation	Unimodal	Gesture	1,143 (29)	7.46 (0.62)
Negation	Unimodal	Speech	1,204 (32)	5.44 (0.61)

Note. RTs in milliseconds, ERs in %. *SEMs* of all aggregated values are included in parentheses.

Table C2

Statistical Results, for the Main RT Analysis in Experiment 3.

Reaction Times			
Effect	Test Statistic (F/t)	p	$p < .05$
Polarity	$F(1, 130) = 165.32$	< .001	***
Compatibility	$F(1.68, 218.57) = 136.02$	< .001	***
Follow-up analysis, comparing Compatibility levels:			
Comp—Uni	$t(130) = 7.03$	< .001	***
Incomp—Uni	$t(130) = 15.27$	< .001	***
Incomp—Comp	$t(130) = 11.38$	< .001	***
There was a significant effect of Compatibility on RTs. Follow-up analyses revealed that all 3 factor levels differed significantly with Unimodal trials eliciting the fastest responses followed by compatible and lastly incompatible trials.			
Compatibility:Polarity	$F(1.84, 238.64) = 21.64$	< .001	***
Follow-up analysis, Polarity effect across Compatibility levels:			
(Comp, Uni):(Aff, Neg)	$F(1, 130) = 0.12$	1.000	
(Incomp, Uni):(Aff, Neg)	$F(1, 130) = 39.44$	< .001	***
(Incomp, Comp):(Aff, Neg)	$F(1, 130) = 23.60$	< .001	***
There was a significant interaction between Compatibility and Polarity. The Follow-up analysis indicates that this interaction was present across all three factor levels of compatibility.			

Note. * < .05, ** < .01, *** < .001, Comp stands for compatible, Incomp for incompatible and Uni for unimodal, Aff stands for affirmative, Neg for negative. The Follow-up analysis of the Compatibility:Polarity consist of three different versions of the interaction, within the parentheses, the (abbreviated) levels of each factor for the investigated interaction are listed.

Table C3

Statistical Results, for the Main ER Analysis in Experiment 3.

Error Rates			
Effect	Test Statistic (*F*/*t*)	*p*	*p* < .05
Polarity	*F*(1, 130) = 8.61	.004	**
Compatibility	*F*(1.91, 248.26) = 39.40	< .001	***
Follow-up analysis, comparing Compatibility levels:			
Comp—Uni	*t*(130) = 0.38	1.000	
Incomp—Uni	*t*(130) = 7.96	< .001	***
Incomp—Comp	*t*(130) = 6.86	< .001	***
There was a significant effect of Compatibility on ERs. Follow-up analyses revealed no evidence that unimodal and compatible trials differed. There was however evidence that the incompatible condition differed significantly from both unimodal and compatible trials.			
Compatibility:Polarity	*F*(1.94, 252.72) = 7.24	< .001	***
Follow-up analysis, Polarity effect across Compatibility levels:			
(Comp, Uni):(Aff, Neg)	*F*(1, 130) = 0.06	1.000	
(Incomp, Uni):(Aff, Neg)	*F*(1, 130) = 10.01	.006	**
(Incomp, Comp):(Aff, Neg)	*F*(1, 130) = 10.32	.005	**
There was a significant interaction of compatibility and polarity on ERs. Follow-up analyses revealed that this effect was driven by the incompatible trials. There was no evidence for an interaction across compatible and unimodal trials only.			

Note. * < .05, ** < .01, *** < .001, Comp stands for compatible, Incomp for incompatible and Uni for unimodal, Aff stands for affirmative, Neg for negative. The Follow-up analysis of the Compatibility:Polarity consist of three different versions of the interaction, within the parentheses, the (abbreviated) levels of each factor for the investigated interaction are listed.

Table C4

Statistical Results, Additional analysis on the Influence of Modality in Experiment 3 (Part I).

Reaction Times			
Effect	Test Statistic (F/t)	p	$p < .05$
Modality	$F(1, 130) = 0.00$	.991	
Polarity	$F(1, 130) = 163.38$	< .001	***
Compatibility	$F(1.71, 222.74) = 135.72$	< .001	***
Follow-up analysis, comparing Compatibility levels:			
Comp—Uni	$t(130) = 7.05$	< .001	***
Incomp—Uni	$t(130) = 15.33$	< .001	***
Incomp—Comp	$t(130) = 11.14$	< .001	***
There was a significant effect of Compatibility on RTs. Follow-up analyses revealed that all 3 factor levels differed significantly with Unimodal trials eliciting the fastest responses followed by compatible and lastly incompatible trials.			
Compatibility:Polarity	$F(1.82, 236.38) = 21.21$	< .001	***
Follow-up analysis, Polarity effect across Compatibility levels:			
(Comp, Uni):(Aff, Neg)	$F(1, 130) = 0.10$	1.000	
(Incomp, Uni):(Aff, Neg)	$F(1, 130) = 39.96$	< .001	***
(Incomp, Comp):(Aff, Neg)	$F(1, 130) = 22.88$	< .001	***
There was a significant interaction of Compatibility and Polarity on RTs. Follow-up analyses revealed that this interaction was driven by incompatible trials as it did not reach significance between compatible and unimodal trials only.			

Note. * < .05, ** < .01, *** < .001, Comp stands for compatible, Incomp for incompatible and Uni for unimodal, Aff stands for affirmative, Neg for negative and Gest stands for gesture. The Follow-up analysis of the Compatibility:Polarity consist of three different versions of the interaction, within the parentheses, the (abbreviated) levels of each factor for the investigated interaction are listed.

Table C5

Statistical Results, Additional analysis on the Influence of Modality in Experiment 3 (Part II).

Reaction Times			
Effect	Test Statistic (F/t)	p	$p < .05$
Compatibility:Modality	$F(1.61, 208.84) = 60.28$	< .001	***
Follow-up analysis, Modality effect across Compatibility levels:			
(Comp, Uni):(Gest, Speech)	$F(1, 130) = 108.63$	< .001	***
(Incomp, Uni):(Gest, Speech)	$F(1, 130) = 42.52$	< .001	***
(Incomp, Comp):(Gest, Speech)	$F(1, 130) = 14.93$	< .001	***
There was a significant interaction of Compatibility and Modality on RTs. Follow-up analyses revealed that this interaction was present across all three factor levels of compatibility.			
Polarity:Modality	$F(1, 130) = 7.95$	.006	**
Follow-up analysis, Polarity grouped by Modality:			
Gesture	$t(130) = 12.16$	< .001	***
Speech	$t(130) = 9.71$	< .001	***
There was a significant interaction of Polarity and Modality on RTs. Follow-up analyses revealed a larger effect of Polarity on gestural compared to verbal trials.			
Compatibility:Polarity:Modality	$F(1.76, 228.65) = 3.12$	.053	
Follow-up analysis, Compatibility:Polarity Interaction across Modality levels:			
Gesture (Comp:Pola)	$F(2, 130) = 16.49$	< .001	***
Speech (Comp:Pola)	$F(2, 130) = 10.35$	< .001	***
There was no evidence for a significant three-way interaction between Compatibility, Polarity and Modality.			

Note. * < .05, ** < .01, *** < .001, Comp stands for compatible, Incomp for incompatible and Uni for unimodal, Aff stands for affirmative, Neg for negative and Gest stands for gesture. The Follow-up analysis of the Compatibility:Polarity consist of three different versions of the interaction, within the parentheses, the (abbreviated) levels of each factor for the investigated interaction are listed.

Appendix D

Stimuli

Table D1

Onset of Speech and Gesture for each Video, Grouped by Actor × Speech Polarity × Gesture Polarity

Speech	Gesture	Gesture Type	Speech Onset	Gesture Onset
Female				
Affirmation	Affirmation	Head	200	183 (11)
Affirmation	Affirmation	Thumb	211	233 (14)
Affirmation	Negation	Head	240	217 (13)
Affirmation	Negation	Thumb	272	233 (14)
Negation	Affirmation	Head	214	200 (12)
Negation	Affirmation	Thumb	250	233 (14)
Negation	Negation	Head	230	200 (12)
Negation	Negation	Thumb	214	250 (15)
Male				
Affirmation	Affirmation	Head	241	200 (12)
Affirmation	Affirmation	Thumb	234	217 (13)
Affirmation	Negation	Head	259	183 (11)
Affirmation	Negation	Thumb	252	233 (14)
Negation	Affirmation	Head	221	233 (14)
Negation	Affirmation	Thumb	229	183 (11)
Negation	Negation	Head	195	167 (10)
Negation	Negation	Thumb	207	217 (13)

Note. All values are in ms. Since gesture onset was extracted by inspection of the videos, the corresponding frames of onset are provided in parentheses.

Table D2
Peak and Integrated Loudness of the Stimuli Grouped by Actor × Speech Polarity × Gesture Polarity × Gesture Type

Speech	Gesture	Gesture Type	L_{pk} (dB)	L_K (LUFS)
Female				
Affirmation	Affirmation	Head	-25.75	-37.39
Affirmation	Affirmation	Thumb	-26.99	-39.13
Affirmation	Negation	Head	-24.84	-36.09
Affirmation	Negation	Thumb	-28.87	-39.84
Negation	Affirmation	Head	-29.71	-39.25
Negation	Affirmation	Thumb	-24.42	-36.22
Negation	Negation	Head	-26.31	-37.45
Negation	Negation	Thumb	-26.11	-37.79
Male				
Affirmation	Affirmation	Head	-13.35	-25.03
Affirmation	Affirmation	Thumb	-14.64	-27.44
Affirmation	Negation	Head	-16.08	-28.88
Affirmation	Negation	Thumb	-12.71	-24.49
Negation	Affirmation	Head	-18.48	-31.18
Negation	Affirmation	Thumb	-19.63	-32.22
Negation	Negation	Head	-18.33	-31.11
Negation	Negation	Thumb	-16.42	-29.18

Note. L_{pk} = *Peak Loudness (dB)*, L_K = *Integrated Loudness (LUFS).*